ADHD Second Brain:

An On-Device AI-Powered macOS Application for ADHD Productivity and Digital Wellbeing

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Statement of Originality

I hereby certify that the work embodied in this report is the result of original research and has not been submitted for a higher degree to any other University or Institution.

Disclosure of AI Tool Usage

In accordance with NTU's policy on the use of generative AI tools in academic work (effective November 2025), the following AI tools were used during the preparation of this report:

- **Claude (Anthropic)**: code review and literature synthesis.
- **ChatGPT (OpenAI)**: brainstorming and proofreading selected sections.

All AI-generated content was reviewed, verified, and substantially revised by the author. The intellectual contributions, system design, implementation, and evaluation are entirely the author's own work.

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Supervisor Declaration Statement

I have reviewed the content and presentation of this report and confirm that it represents the student's own work, conducted under my supervision.

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Authorship Attribution Statement

This report is written entirely by the undersigned. All work presented in this report was carried out by the author unless otherwise stated. Where the work of others has been used, it has been duly acknowledged and referenced.

The following contributions are noted:

- The SenticNet APIs (Chapter 4, Section 4.4) were provided by the SenticNet research group at NTU.
- The Mem0 memory framework (Chapter 4, Section 4.5) is an open-source library used under the Apache 2.0 licence.
- The Apple MLX framework (Chapter 4, Section 4.2) is an open-source library developed by Apple Inc.

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Abstract

Attention-Deficit/Hyperactivity Disorder (ADHD) affects an estimated 2.58% to 6.76% of the global adult population, imposing a substantial economic burden of approximately USD 4,336 per affected worker per year in lost productivity. The proliferation of digital devices and persistent connectivity has compounded core ADHD symptoms, including attentional dysregulation, working memory deficits, and impaired executive function. Despite the prevalence of productivity applications on the market, no integrated desktop tool existed that combined conversational coaching, real-time emotion awareness, screen activity monitoring, and persistent semantic memory within a single privacy-preserving framework.

This project developed an on-device AI-powered macOS application that addressed these gaps through the integration of several complementary subsystems. A Qwen3-4B large language model, executed locally via the Apple MLX framework, provided conversational coaching at 37.1 tokens per second. A SetFit contrastive learning pipeline achieved 86% emotion classification accuracy across six ADHD-specific affective categories using only 210 training sentences. SenticNet affective computing, accessed through 13 application programming interfaces and grounded in the Hourglass of Emotions model, supplied fine-grained sentiment and sentic features with 100% reliability. Mem0 semantic memory, backed by pgvector, attained 89% top-1 relevance in context retrieval. A five-tier activity classification engine processed screen titles at 549 titles per second, and Just-in-Time Adaptive Interventions, structured around Barkley's executive function domains, delivered context-sensitive behavioural nudges.

Evaluation demonstrated that the system consumed 21,314 millijoules per inference, representing approximately 1.4% battery drain per hour, while maintaining a peak memory footprint below 3 gigabytes. All predefined project objectives were exceeded. The application constituted the first macOS-native ADHD assistant with a fully on-device large language model, addressed eight identified market gaps, and contributed an open-source evaluation framework with reproducible benchmarks.

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List of Abbreviations

Abbreviation	Definition
ADHD	Attention-Deficit/Hyperactivity Disorder
AI	Artificial Intelligence
ANE	Apple Neural Engine
API	Application Programming Interface
ASRS	Adult ADHD Self-Report Scale
CBM	Concept Bottleneck Model
CBT	Cognitive Behavioural Therapy
CLI	Command-Line Interface
CPU	Central Processing Unit
CV	Coefficient of Variation
DRAM	Dynamic Random-Access Memory
E2E	End-to-End
EF	Executive Function
F1	F1-Score (harmonic mean of precision and recall)
GPU	Graphics Processing Unit
HRV	Heart Rate Variability
IEEE	Institute of Electrical and Electronics Engineers
IRB	Institutional Review Board
JITAI	Just-in-Time Adaptive Intervention
LLM	Large Language Model
LoRA	Low-Rank Adaptation
MLX	Apple Machine Learning Framework
MPS	Metal Performance Shaders
NLP	Natural Language Processing
NTU	Nanyang Technological University
PARA	Projects, Areas, Resources, Archives
PKM	Personal Knowledge Management
QLoRA	Quantised Low-Rank Adaptation
RAG	Retrieval-Augmented Generation
RSS	Resident Set Size
STS	Semantic Textual Similarity

Abbreviation	Definition
SUS	System Usability Scale
XAI	Explainable Artificial Intelligence

Chapter 1

Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is among the most prevalent neurodevelopmental conditions worldwide, yet digital interventions for adults with ADHD remain remarkably underdeveloped. This chapter motivates the need for an integrated, privacy-preserving desktop application that combines on-device language modelling, affective computing, screen activity monitoring, and persistent memory to support adults with ADHD in knowledge work. The chapter articulates the specific problem, defines measurable objectives, delineates scope and limitations, and summarises the contributions of this work.

1.1 Background and Motivation

1.1.1 ADHD Prevalence and the Executive Function Deficit

Attention-Deficit/Hyperactivity Disorder is a neurodevelopmental condition characterised not merely by difficulties sustaining attention, but by pervasive impairments in executive function—the set of cognitive processes that enable goal-directed behaviour, working memory, inhibitory control, and cognitive flexibility [1]. Meta-analytic estimates place the global adult prevalence of ADHD between 2.58% and 6.76% [2, 3], translating to over 200 million adults worldwide who navigate daily life with executive function profiles that diverge substantially from the neurotypical baseline upon which most productivity tools and digital interfaces are designed. ADHD in adults is associated with reduced occupational attainment, higher rates of job turnover, and elevated risk of comorbid anxiety and mood disorders, yet the technology ecosystem has been slow to develop tools that accommodate this specific cognitive profile.

1.1.2 Economic Impact and the Knowledge Worker Gap

The economic burden of adult ADHD is substantial. Barkley and Murphy estimated the excess annual cost at $4,336 per worker per year, attributable to reduced productivity, increased absenteeism, and elevated error rates [4]. In the knowledge economy, where

the desktop computer is the primary instrument of production, this cost accrues overwhelmingly at the human-computer interface. Adults with ADHD spend most of their working hours interacting with desktop applications—email clients, web browsers, code editors—yet these applications assume intact working memory, sustained attention, and consistent self-regulation. An application that could reduce even a fraction of this productivity loss would need to understand what the user is doing, how the user is feeling, and what the user has done before, all without sending sensitive data to external servers.

1.1.3 The Digital Distraction Paradox

A particularly vexing challenge is the digital distraction paradox: ADHD predisposes individuals to distraction by digital stimuli, while excessive digital engagement exacerbates attentional difficulties [5, 6]. The desktop computer is simultaneously the primary workspace and the primary distraction vector. Existing approaches—website blockers, application timers, focus modes—treat the symptom rather than the cause, imposing external constraints without understanding the cognitive or affective state that precipitated the distraction. Despite growing interest in digital mental health interventions, the evidence base for ADHD-specific applications remains thin, with a recent meta-analysis finding only a modest effect size of Hedges' $g = -0.32$ [7]. Among ADHD-specific tools, Brain.fm is one of the very few with peer-reviewed evidence [8]; the broader landscape consists overwhelmingly of static task managers and timers that lack adaptive intelligence, affective awareness, or personalised coaching. A more effective intervention would monitor screen activity in real time, detect drift from productive work, and intervene with context-aware coaching that accounts for the user's emotional state and historical patterns.

1.1.4 Emotion Dysregulation as a Core Symptom

Recent clinical research has increasingly recognised emotion dysregulation not as a mere comorbidity, but as a fourth core symptom of ADHD alongside inattention, hyperactivity, and impulsivity [9]. Adults with ADHD frequently experience rapid, intense emotional responses—frustration when tasks prove difficult, anxiety when deadlines loom, and overwhelming cognitive overload when demands exceed executive capacity. These affective states are primary drivers of task abandonment, not incidental to it. Any comprehensive support system must therefore incorporate affective awareness: an application that detects frustration or anxiety and responds with appropriately timed, empathetic coaching has the potential to interrupt the emotion–distraction–guilt cycle

that characterises much of the ADHD experience in knowledge work.

1.1.5 The Second Brain Paradigm and the Privacy Imperative

The "Second Brain" paradigm, popularised by Forte [10], advocates for the systematic externalisation of knowledge into a persistent digital system that augments human memory. This paradigm is especially relevant for adults with ADHD, whose working memory deficits mean that valuable insights and contextual knowledge are routinely lost between sessions. However, the tools most commonly associated with this paradigm—Notion, Obsidian, Roam Research—require precisely the executive functions that ADHD impairs: consistent categorisation, regular review, and disciplined capture habits. A truly ADHD-appropriate Second Brain must automate the capture and organisation process, requiring minimal executive overhead from the user.

Such a system produces data of extraordinary sensitivity—screen content, emotional states, conversational transcripts—creating unacceptable privacy risks if transmitted to cloud servers. Recent advances in edge AI have made on-device processing a viable alternative. The MLX framework, developed by Apple Machine Learning Research, enables efficient neural network inference on Apple Silicon hardware [11], and models such as Qwen3-4B [12] achieve competitive performance while fitting within the memory constraints of consumer hardware. Together, these technologies make it feasible to run a capable language model, an emotion classifier, and a memory system entirely on the user's device.

1.2 Problem Statement

Adults with ADHD who work primarily on desktop computers face a fragmented landscape of support tools, typically requiring three to five separate applications: a task manager, a focus timer, a note-taking system, a calendar, and possibly a mood tracker. Each operates in isolation, unaware of the user's activity in other applications, emotional state, or historical patterns. This fragmentation is itself an ADHD tax—the cognitive overhead of switching between tools actively undermines the executive functions the tools are ostensibly designed to support.

From a technical standpoint, no existing application integrates on-device large language model coaching, real-time affective computing, continuous screen activity monitoring, and persistent conversational memory within a single desktop application. Cloud-based solutions exist for individual components but introduce latency, require network

connectivity, and create the privacy vulnerabilities discussed in section 1.1.5. The technical challenge is to unify these capabilities into a coherent, responsive, and private desktop application that runs entirely on consumer hardware.

1.3 Project Objectives

This project aims to design, implement, and evaluate a desktop application—the ADHD Second Brain—that provides integrated, privacy-preserving cognitive support for adults with ADHD. The application targets macOS on Apple Silicon hardware, leveraging the MLX framework for on-device inference. Five specific, measurable objectives guide the project.

Objective 1: On-Device LLM Coaching. The application shall provide real-time conversational coaching through Qwen3-4B on the MLX framework, targeting at least 30 tokens per second. The achieved throughput of 37.1 tok/s exceeds the target by 23.7%.

Objective 2: Emotion Classification. The application shall classify user-expressed emotion into six ADHD-relevant categories with at least 80% accuracy. The achieved accuracy of 86% (macro-F1 0.862), trained on only 210 sentences, demonstrates that few-shot contrastive learning yields clinically useful classifiers.

Objective 3: Screen Activity Classification. The application shall classify active window titles in real time. The achieved throughput of 549 titles per second, with 78% resolved by deterministic rules, ensures negligible computational overhead.

Objective 4: Persistent Conversational Memory. The application shall maintain persistent memory using the Mem0 framework, targeting at least 90% top-1 relevance. The achieved 89% top-1 hit rate ensures contextually grounded coaching responses.

Objective 5: Privacy-Preserving, Energy-Efficient Architecture. All sensitive data shall be processed on-device. The achieved energy consumption of approximately 1.4% battery per hour and peak AI memory below 3 GB confirm suitability for sustained daily use.

Table 1.1 summarises the quantitative targets and achieved results for each objective.

Table 1.1: Summary of project objectives, targets, and achieved results.

Objective	**Target**	**Achieved**
LLM throughput	$\geq$30 tok/s	37.1 tok/s
Emotion accuracy	$\geq$80%	86%
Screen classification	Real-time	549 titles/s
Memory relevance	$\geq$90% top-1	89% top-1
Energy consumption	$<$2% battery/hr	$\sim$1.4% battery/hr
Peak AI memory	$<$6 GB	$<$3 GB

1.4 Project Scope and Limitations

The scope of this project encompasses the design, implementation, and evaluation of a single-user macOS desktop application targeting Apple Silicon M-series processors. The application is implemented in Python with a Swift-based menu bar interface and comprises five core modules: an on-device LLM coaching engine, an emotion classification pipeline, a screen activity monitor and classifier, a persistent memory system, and a just-in-time adaptive intervention (JITAI) orchestrator. The implementation is accompanied by a comprehensive test suite of over 330 unit, integration, and end-to-end tests. All natural language processing is conducted in English, and the system is designed for individual rather than collaborative use.

Several deliberate limitations bound the project. First, the application targets macOS exclusively, as the MLX framework and Apple Silicon GPU acceleration are platform-specific. Second, emotion detection relies solely on text-based classification of user chat messages; multimodal signals are not incorporated. Third, the project does not include an IRB-approved clinical trial with ADHD-diagnosed participants; evaluation is conducted through automated benchmarks and system-level performance metrics. Fourth, the on-device LLM uses prompt engineering rather than parameter-efficient fine-tuning. Fifth, the emotion classifier is trained on expert-labelled synthetic data rather than clinical data collected from individuals with diagnosed ADHD.

1.5 Contributions and Significance

The primary contribution of this project is the design and implementation of the first integrated desktop application that combines on-device large language model coaching, real-time affective computing, continuous screen activity monitoring, and persistent conversational memory for adults with ADHD. While individual components exist as separate tools or cloud services, no prior work has unified these capabilities into

a single, privacy-preserving desktop application optimised for the ADHD cognitive profile. This integration is a clinical necessity, as the fragmentation of support tools imposes precisely the executive overhead that adults with ADHD can least afford.

The project makes four specific technical contributions. First, a four-tier SenticNet integration pipeline that combines API-based affective knowledge with local caching to enrich coaching responses with validated emotional context. Second, a few-shot emotion classification system achieving 86% accuracy across six ADHD-relevant categories using only 210 labelled sentences, demonstrating that contrastive learning with sentence transformers [13] can yield clinically useful classifiers without large-scale annotation. Third, a hybrid screen activity classification system processing 549 titles per second, resolving 78% through deterministic rules and falling back to a 14-class ML classifier with 96.5% accuracy. Fourth, a persistent memory system achieving 89% top-1 retrieval relevance. A systematic analysis of nine existing ADHD and productivity applications identifies eight distinct capability gaps that the ADHD Second Brain addresses, positioning it as a fundamentally different category of application.

The broader significance lies in demonstrating that edge AI is viable for mental health applications. By running a 4-billion-parameter language model, an emotion classifier, and a memory system on a consumer laptop with less than 3 GB peak AI memory and approximately 1.4% battery drain per hour, the project establishes that privacy-preserving, always-on cognitive support is technically feasible on current hardware, providing a reference point for any application seeking to deploy sensitive AI capabilities on-device.

1.6 Report Organisation

The remainder of this report is organised as follows. Chapter 2 reviews the relevant literature on ADHD, digital interventions, affective computing, on-device language models, and memory-augmented AI systems, establishing the theoretical and technical foundations for the system design. Chapter 3 presents the system architecture, detailing the design of each module and the rationale for key architectural decisions. Chapter 4 describes implementation of each component, including emotion classification pipeline, LLM integration, screen monitoring system, and memory framework. Chapter 5 reports the testing methodology and evaluation results, covering unit and integration testing, benchmark performance, and comparative analysis against project objectives. Chapter 6 concludes the report with a summary of achievements, a discussion of limitations, directions for future work, and lessons learned during development.

Chapter 2

Literature Review

This chapter critically examines the existing body of research across the multidisciplinary domains that inform the design of ADHD Brain, an on-device AI-powered macOS application for ADHD productivity and wellbeing. The review is organised thematically, beginning with the clinical foundations of ADHD and executive dysfunction in digital environments, proceeding through technology-based interventions and AI-driven mental health tools, and culminating in an analysis of existing ADHD applications and the identification of research gaps that this project addresses.

2.1 ADHD and Executive Function in Digital Environments

2.1.1 Prevalence and Clinical Significance of Adult ADHD

Attention-Deficit/Hyperactivity Disorder (ADHD) is among the most extensively studied neurodevelopmental conditions, yet its prevalence in adulthood remains widely underappreciated. The World Federation of ADHD International Consensus Statement, authored by Faraone et al. [2] and endorsed by over 80 leading researchers, established that ADHD persists into adulthood at rates ranging from 2.58% (narrow, persistent diagnostic criteria) to 6.76% (symptomatic but potentially sub-threshold), thereby dispelling the long-held misconception that ADHD is exclusively a childhood disorder. This finding is corroborated by Song et al. [3], whose global meta-analysis estimated that between 139 million and 366 million adults worldwide live with ADHD, depending on the diagnostic threshold applied. The paediatric literature further contextualises the developmental trajectory: Salari et al. [14] reported a pooled prevalence of approximately 7.6% among children and adolescents, a substantial proportion of whom will carry clinically significant symptoms into adulthood. Taken together, these epidemiological findings underscore that any productivity or wellbeing tool targeting adult populations must account for the high base rate of ADHD and the heterogeneity of its presentation across the lifespan.

2.1.2 Executive Function Deficits as the Core Mechanism

The functional impairments associated with ADHD are best understood through the lens of executive function (EF) theory. Diamond [1] identified three core executive functions—inhibitory control (including self-regulation), working memory, and cognitive flexibility—and demonstrated that these foundational capacities underpin higher-order skills such as planning, reasoning, and problem-solving. In the context of ADHD, deficits in inhibitory control manifest as difficulty resisting distractions, working memory impairments result in the loss of task-relevant information mid-activity, and reduced cognitive flexibility hinders the ability to shift between tasks adaptively. These deficits carry substantial economic consequences: Barkley and Fischer [4] estimated an occupational impairment cost of $4,336 per worker per year attributable to ADHD-related productivity losses, encompassing absenteeism, presenteeism, and elevated turnover. This economic dimension provides a compelling justification for tools that directly scaffold executive function rather than merely providing generic productivity features.

2.1.3 Digital Environments as Amplifiers of ADHD Symptoms

The proliferation of digital devices and always-connected workflows introduces unique challenges for individuals with executive function deficits. Thorell et al. [5] conducted a longitudinal investigation into the association between digital media use and ADHD symptoms, finding that the relationship is bidirectional: ADHD predisposes individuals to excessive and poorly regulated digital media consumption, while prolonged digital engagement further exacerbates attentional difficulties. Dontre [6] extended this line of inquiry to academic contexts, demonstrating that technology-mediated distraction significantly impairs learning outcomes, with the magnitude of impairment modulated by individual differences in self-regulation capacity—precisely the capacity that is diminished in ADHD.

Most critically for the present work, Beattie [9] investigated the phenomenon of digital disconnection in adults with ADHD and proposed that emotion dysregulation should be recognised as a "fourth core symptom" alongside inattention, hyperactivity, and impulsivity. This reconceptualisation is significant because it implies that effective ADHD tools must address not only cognitive scaffolding (task management, distraction blocking) but also affective monitoring and regulation. The design of ADHD Brain directly incorporates this insight through its emotion classification pipeline and adaptive intervention system, which together aim to detect and respond to emotional states that precede productivity collapse.

2.2 Technology-Based ADHD Interventions

2.2.1 Evidence Base for Digital ADHD Interventions

The efficacy of technology-based interventions for ADHD has been the subject of increasing systematic scrutiny. Garcia-Peral et al. [7] conducted an umbrella review synthesising 26 systematic reviews encompassing 34,442 participants, representing the most comprehensive meta-analytic assessment of digital ADHD interventions to date. The pooled effect size of Hedges' $g = -0.32$ indicates a small-to-moderate treatment effect, which, while statistically significant, falls considerably short of the effect sizes reported for pharmacological interventions. This modest aggregate effect conceals substantial heterogeneity, however: interventions that were personalised, multi-component, and sustained over time consistently outperformed static, one-size-fits-all approaches. This finding directly informs the architectural decision in ADHD Brain to employ adaptive, multi-tier interventions that escalate in intensity based on real-time affective and behavioural signals rather than relying on fixed schedules or user-initiated triggers.

2.2.2 Population-Specific Evidence and Gaps

The evidence base is unevenly distributed across age groups and intervention modalities. Shou et al. [15] conducted a digital therapy meta-analysis focused specifically on children with ADHD, finding promising effects for gamified cognitive training but limited evidence for sustained transfer to real-world executive function. Wong et al. [16] performed a randomised controlled trial (RCT) meta-analysis restricted to school-age populations, confirming moderate effects on attention but negligible effects on hyperactivity-impulsivity, suggesting that digital interventions may differentially address specific symptom clusters. For adults—the target population of the present work—the evidence is notably thinner. Elbe et al. [17] reviewed computerised cognitive interventions for adults with ADHD and concluded that while some promise exists, the field suffers from small sample sizes, short follow-up periods, and inconsistent outcome measures. The authors called for interventions that integrate into daily workflows rather than requiring dedicated training sessions, a principle that ADHD Brain operationalises through its ambient monitoring approach.

2.2.3 Design Principles for Digital Mental Health Interventions

Beyond efficacy data, the design of digital interventions is itself an active area of research. Storetvedt et al. [18] proposed a framework for digital mental health intervention design that emphasises user agency, contextual sensitivity, and iterative personalisation. Their framework highlights the tension between automation (which reduces the executive function burden on users) and user control (which preserves autonomy and reduces reactance). This tension is particularly acute for ADHD users, who simultaneously need more automation due to executive function deficits and are more sensitive to perceived loss of control. The design of ADHD Brain navigates this tension by providing automated detection and suggestion while preserving user veto authority at every intervention tier, a design philosophy that draws directly from Storetvedt et al.'s recommendations.

2.3 AI and LLM-Powered Mental Health Applications

2.3.1 Conversational AI: Foundational Evidence

The deployment of artificial intelligence in mental health contexts has evolved rapidly from rule-based chatbots to sophisticated large language model (LLM) systems. Fitzpatrick et al. [19] provided early and rigorous evidence for the efficacy of conversational AI through their RCT of Woebot, a cognitive-behavioural therapy (CBT) chatbot. Participants in the Woebot condition demonstrated significantly reduced depression symptoms compared to an information-only control group ($F = 6.47$, $P = .01$), with engagement rates exceeding those typical of digital mental health tools. This study established that automated conversational agents could deliver therapeutic content with measurable clinical benefit, though it also highlighted that Woebot's rule-based architecture limited its ability to handle off-script interactions—a limitation that LLM-based systems aim to overcome.

2.3.2 The LLM Revolution in Mental Health Chatbots

The advent of large language models has fundamentally expanded the capabilities of mental health chatbots. Yuan et al. [20] conducted a comprehensive review identifying over 50 mental health chatbot systems, of which 22 employed LLM architectures. The authors observed that LLM-based systems demonstrated superior naturalistic conversation, reduced user frustration with scripted responses, and enabled more flexible therapeutic dialogue. However, they also identified critical risks, including hallucina-

tion of clinical information, inconsistent adherence to therapeutic protocols, and the absence of robust safety guardrails. Amin et al. [21] corroborated these findings in a scoping review of AI chatbots for mental health, noting that while user satisfaction was generally high, clinical validation remained sparse and regulatory frameworks were virtually nonexistent. The convergence of strong user demand with inadequate safety infrastructure presents both an opportunity and a responsibility for systems such as ADHD Brain, which must balance conversational capability with principled clinical boundaries.

2.3.3 Prompt Engineering and Safety Frameworks

The risks identified in the chatbot literature have catalysed research into safety-oriented prompt engineering. Boit et al. [22] proposed the MIND-SAFE framework, a structured approach to prompt engineering for mental health LLM applications that enforces clinical boundaries, escalation protocols, and harm-reduction principles at the prompt level rather than relying solely on model fine-tuning. This approach is particularly relevant for on-device deployments where smaller models may exhibit less reliable safety behaviour than their larger cloud-based counterparts. Sharma et al. [23] complemented this work by studying user experiences with LLM chatbots in mental health contexts, finding that users valued empathetic tone, contextual memory, and the ability to revisit previous conversations—features that informed the memory architecture of ADHD Brain. The present system incorporates structured prompt engineering with safety boundaries while leveraging Mem0-based memory to provide the conversational continuity that users identify as critical to therapeutic alliance.

2.4 Affective Computing and SenticNet

2.4.1 The Evolution of SenticNet

Affective computing—the computational detection, interpretation, and simulation of human emotion—represents a critical enabling technology for any system that aims to respond to user emotional states. The SenticNet knowledge base, developed by Cambria and colleagues over nearly 15 years [24, 25, 26, 27, 28], represents one of the most sustained efforts to build a computationally tractable representation of common-sense affective knowledge. From its inception as a semantic network mapping concepts to affective valence (SenticNet 1), the system has evolved through successive versions to incorporate polarity detection (SenticNet 2), conceptual primitives (SenticNet 3), an affective algebra (SenticNet 4), a multidimensional sentiment analysis framework (Sen-

ticNet 5), top-down and bottom-up attention mechanisms (SenticNet 6), neurosymbolic reasoning (SenticNet 7), and most recently, large-scale integration with transformer architectures (SenticNet 8).

The theoretical foundation underpinning SenticNet is the Hourglass of Emotions model, which decomposes affective states into four independent dimensions—pleasantness, attention, sensitivity, and aptitude—each modelled as a continuous value between -1 and $+1$. Cambria [25] situated this progression within the broader discipline of affective computing and sentiment analysis, arguing that robust affect detection requires the fusion of statistical learning with structured common-sense knowledge. The institutional connection to Nanyang Technological University, where Cambria's research group is based, provides a natural point of intersection with the present work, and the initial design of ADHD Brain drew heavily on SenticNet as an affective feature source.

2.4.2 SenticNet in the Context of ADHD Emotion Classification

The application of SenticNet to ADHD-specific emotion classification presents both opportunities and challenges that were empirically investigated during the development of ADHD Brain. The initial design hypothesis was that SenticNet's concept-level affective features could augment embedding-based classifiers by providing structured affective priors. Specifically, the hybrid approach (Approach A in the system's classifier evaluation) combined sentence embeddings from pre-trained transformer models [13, 29] with SenticNet-derived features including pleasantness, attention, sensitivity, and aptitude dimensions. However, empirical evaluation revealed that SenticNet features did not improve classification accuracy for the ADHD-specific emotion taxonomy (six categories: joyful, focused, frustrated, anxious, disengaged, overwhelmed). The embedding-only variant achieved 74% accuracy while the hybrid variant incorporating SenticNet features achieved only 70%, indicating that SenticNet's general-purpose affective dimensions introduced noise when applied to a domain-specific affective taxonomy. This finding aligns with the broader critique of general-purpose sentiment lexicons: while SenticNet excels at binary sentiment classification and dimensional affect modelling, its concept-level granularity does not naturally map to the nuanced, context-dependent emotional states characteristic of ADHD—for example, the distinction between "anxious" and "overwhelmed," which share similar valence and arousal profiles but differ in their functional implications for intervention design.

2.4.3 Beyond SenticNet: Contrastive and Interpretable Approaches

The limitations of SenticNet-augmented classification motivated exploration of alternative approaches that could achieve higher accuracy with limited labelled data while maintaining interpretability. The contrastive learning approach (Approach B), inspired by the SetFit methodology but implemented independently due to library incompatibilities with current transformer versions, achieved 86% accuracy using CoSENT loss with the all-mpnet-base-v2 sentence transformer [13]. This result demonstrates that for domain-specific affective classification with limited training data (210 curated sentences), contrastive representation learning substantially outperforms both lexicon-augmented and standard fine-tuning approaches. To address the interpretability requirements of an affective computing system—particularly one operating in a mental health-adjacent context—the system additionally draws on the concept bottleneck model (CBM) paradigm [30, 31] and Testing with Concept Activation Vectors (TCAV) [32]. These interpretability frameworks enable the system to provide human-understandable explanations for emotion classifications, a critical property for user trust and for enabling users to correct misclassifications that would otherwise lead to inappropriate interventions.

2.5 Second Brain and Personal Knowledge Management

2.5.1 The Second Brain Paradigm

The "second brain" concept, popularised by Forte [10], proposes a systematic approach to personal knowledge management (PKM) built on two complementary frameworks: the PARA organisational system (Projects, Areas, Resources, Archives) and the CODE methodology (Capture, Organise, Distil, Express). The central argument is that externalising knowledge into a trusted, searchable system frees cognitive resources for creative and analytical work—a proposition with particular resonance for individuals with ADHD, whose working memory deficits make internal knowledge retention unreliable. The act of capture alone reduces the cognitive load associated with maintaining mental inventories of tasks, ideas, and commitments, which is precisely the type of sustained working memory operation that individuals with ADHD find most taxing [1]. However, Forte's framework assumes a level of sustained organisational effort that is itself dependent on the executive functions impaired in ADHD, creating a fundamental paradox: those who would benefit most from a second brain are least equipped to main-

tain one manually. This paradox motivates the integration of AI-assisted automation into the knowledge management pipeline, a design decision that is central to ADHD Brain.

2.5.2 PKM in Practice and the Role of AI

Empirical research on PKM practices illuminates both the promise and the pitfalls of the second brain approach. Da Silva et al. [33] investigated the relationship between PKM and self-directed learning, finding that effective PKM correlates with improved learning outcomes but requires metacognitive skills that are unevenly distributed in the population. Ferreira et al. [34] conducted a qualitative study of how people actually manage knowledge in second brain systems, revealing that most users struggle with the "organise" and "distil" phases, leading to knowledge graveyards of captured but never-processed information. Razmerita et al. [35] provided an earlier theoretical grounding for PKM in the context of Web 2.0 technologies, arguing that social and technological affordances must align with individual cognitive styles. Most recently, the ACM Group [36] explored the emerging role of AI companions in PKM systems, finding that AI-assisted capture, tagging, and retrieval substantially reduced the organisational burden that constitutes the primary barrier to sustained PKM practice. ADHD Brain operationalises this insight by automating the capture and organisation phases through AI-powered processing of user inputs, thereby reducing the executive function demands that cause conventional second brain systems to fail for ADHD users.

2.6 On-Device AI and Privacy-Preserving Inference

2.6.1 The Case for Edge Deployment

The deployment of LLMs on edge devices, rather than via cloud APIs, represents a critical architectural decision with implications for privacy, latency, cost, and accessibility. Lin et al. [37] provided a comprehensive survey of edge LLM deployment in *ACM Computing Surveys*, cataloguing the technical challenges—memory constraints, thermal throttling, energy consumption—and the emerging solutions—quantisation, pruning, knowledge distillation, and speculative decoding—that make local inference increasingly viable. Xu et al. [38] surveyed the broader mobile edge intelligence landscape, situating LLM deployment within a continuum of on-device AI capabilities and arguing that the convergence of model compression techniques and hardware acceleration is approaching a tipping point at which meaningful language model inference becomes feasible on consumer devices. For mental health applications, the privacy implications

of edge deployment are particularly salient: users are far more willing to disclose sensitive emotional and behavioural information to a system that processes data locally than to one that transmits data to cloud servers [23]. This privacy advantage is not merely a matter of user preference but carries clinical significance, as willingness to disclose directly affects the quality and completeness of the data available to the system for personalisation.

2.6.2 Apple Silicon as an Inference Platform

The Apple Silicon architecture, with its unified memory design and dedicated Neural Engine, has emerged as a particularly capable platform for on-device LLM inference. Hannun et al. [11] introduced MLX, Apple's open-source machine learning framework optimised for Apple Silicon, which provides a NumPy-like API with automatic differentiation and GPU acceleration via the Metal backend. Ajayi et al. [39] benchmarked LLM inference performance on Apple Silicon, demonstrating that M-series chips achieve competitive throughput relative to discrete GPUs for models up to approximately 8 billion parameters, with the unified memory architecture eliminating the CPU-GPU transfer bottleneck that constrains traditional inference pipelines. Barrios et al. [40] extended this analysis to native LLM inference workloads, confirming that quantised models (4-bit and 8-bit) running on Apple Silicon can achieve interactive token generation rates (>20 tokens/second) for models in the 3B–8B parameter range. Apple's own technical report on Apple Intelligence [41] described the foundation models deployed across iPhone, iPad, and Mac, demonstrating the company's investment in on-device AI infrastructure. Cai et al. [42] provided the most recent analysis of efficient inference techniques for edge LLMs, cataloguing advances in KV-cache compression, speculative decoding, and mixed-precision execution that collectively push the frontier of what is achievable on consumer hardware. The convergence of these developments establishes Apple Silicon as a viable deployment target for the interactive, privacy-preserving LLM inference that ADHD Brain requires.

2.6.3 Model Selection for On-Device ADHD Support

The choice of base model for on-device deployment involves navigating trade-offs between capability, latency, and memory footprint. Yang et al. [12] introduced the Qwen3 model family, which spans dense architectures from 0.6B to 32B parameters and mixture-of-experts (MoE) architectures from 30B to 235B parameters. For the on-device deployment context of ADHD Brain, the smaller dense variants (0.6B–8B) are most relevant, as they fit within the memory constraints of consumer Mac hardware

while still demonstrating strong performance on reasoning, instruction-following, and conversational tasks. To further adapt these models to the ADHD domain, parameter-efficient fine-tuning techniques such as LoRA [43] and QLoRA [44] enable domain adaptation without the prohibitive memory and compute requirements of full fine-tuning. LoRA achieves this by decomposing weight updates into low-rank matrices, reducing the number of trainable parameters by several orders of magnitude while preserving model quality, and QLoRA extends this approach by operating on quantised base weights, further reducing the memory footprint during fine-tuning. Chhikara et al. [45] demonstrated that memory-augmented LLM systems using Mem0 achieve a 26% improvement over baseline OpenAI models with 91% lower latency and 90% token savings, providing strong empirical support for the memory architecture adopted in ADHD Brain. The combination of a compact, quantised base model with parameter-efficient fine-tuning and memory augmentation constitutes the inference stack that enables ADHD Brain to deliver personalised, context-aware conversational support entirely on-device.

2.7 Review of Existing ADHD Applications

2.7.1 The Current Application Landscape

The market for ADHD-oriented productivity and wellbeing applications has grown substantially in recent years, yet the landscape remains fragmented, with individual applications addressing narrow slices of the ADHD experience. To contextualise the contribution of ADHD Brain, a systematic competitive analysis was conducted across nine representative applications spanning four functional categories: focus enhancement (Focusmate, Brain.fm), distraction management (Freedom, Cold Turkey), task and routine management (Tiimo, Structured, Routinery), and ADHD-specific support (Inflow, Goblin Tools). The analysis evaluated each application against eight criteria derived from the literature review: integrated macOS support, on-device LLM capability, affective computing, automated distraction detection, personalised memory, multi-tier intervention, academic validation, and cross-functional integration. These criteria were selected because each corresponds to a capability that the literature identifies as important for effective ADHD support but that remains unaddressed or only partially addressed by existing tools.

2.7.2 Comparative Analysis

Table 2.1 presents the results of the competitive analysis across all nine applications and the proposed ADHD Brain system. The table reveals a striking pattern: no existing application satisfies more than two of the eight evaluation criteria, and the majority satisfy none. This finding is consistent across all four functional categories, indicating that the gap is not specific to any one type of application but rather reflects a systemic failure to integrate capabilities that the literature identifies as jointly necessary for effective ADHD support.

Table 2.1: Comparative analysis of existing ADHD applications against eight capability criteria. ✓ = fully supported, ✗ = not supported. ADHD Brain is the system proposed in this work.

Application	**macOS**	**On-Device LLM**	**Affective**	**Auto-Detect**	**Memory**	**Multi-Tier**	**Validated**	**Integrated**
Focusmate	Web	✗	✗	✗	✗	✗	✗	✗
Freedom	✓	✗	✗	Manual	✗	✗	✗	✗
Cold Turkey	✓	✗	✗	Manual	✗	✗	✗	✗
Brain.fm	✓	✗	✗	✗	✗	✗	✓	✗
Inflow	Mobile	✗	✗	✗	Partial	Partial	✗	✗
Tiimo	Mobile	✗	✗	✗	✗	✗	✗	✗
Structured	✓	✗	✗	✗	✗	✗	✗	✗
Goblin Tools	Web	Cloud	✗	✗	✗	✗	✗	✗
Routinery	Mobile	✗	✗	✗	✗	✗	✗	✗
ADHD Brain	✓	✓	✓	**Auto**	✓	✓	**In Prog.**	✓

2.7.3 Critical Analysis of Gaps

The competitive analysis reveals eight significant gaps in the existing landscape that collectively define the opportunity space for ADHD Brain.

First, no current application provides a fully integrated macOS-native solution that combines productivity tools with wellbeing support; users must cobble together multiple applications (e.g., Freedom for distraction blocking, Tiimo for routines, Inflow for psychoeducation), incurring fragmentation overhead that is itself an executive function burden [1]. Second, no existing application employs on-device LLM inference; Goblin Tools utilises cloud-based language models but transmits user data to external servers, which is problematic for sensitive mental health and productivity data. Third, affective computing is entirely absent from the competitive landscape—no application monitors or responds to user emotional states, despite the strong evidence that emotion dysreg-

ulation is a core component of ADHD [9]. Fourth, distraction detection in existing applications (Freedom, Cold Turkey) is exclusively manual, requiring users to predefine blocked sites and applications, which fails to account for context-dependent distraction patterns and imposes the very executive function demands it purports to alleviate.

Fifth, personalised memory—the ability to learn from and adapt to individual user patterns over time—is at best partially implemented (Inflow offers some personalisation of psychoeducational content) but nowhere approaches the semantic memory capabilities enabled by embedding-based retrieval systems. Sixth, multi-tier intervention—the graduated escalation from gentle nudges to assertive interruptions based on real-time assessment—is absent, with existing applications offering only binary states (blocked/unblocked, on-task/off-task). Seventh, with the sole exception of Brain.fm, which Woods et al. [8] validated in a peer-reviewed study published in *Communications Biology*, none of the reviewed applications have undergone rigorous academic evaluation, raising questions about the evidence base on which users are making adoption decisions.

Eighth, and perhaps most consequentially, the requirement to use multiple disconnected applications imposes a coordination cost that disproportionately burdens individuals with ADHD. Each additional application in a user's productivity stack introduces its own interface conventions, notification systems, and configuration requirements, collectively creating an executive function tax that scales with the number of tools employed. The irony is acute: the population that most needs integrated support is forced to perform the most cognitively demanding integration work. ADHD Brain addresses this gap by consolidating distraction management, affective monitoring, conversational AI support, knowledge management, and adaptive intervention into a single, unified application.

2.7.4 Digital Wellbeing Tool Effectiveness

The broader digital wellbeing literature provides additional context for evaluating existing ADHD applications and for informing the design of new ones. Monge Roffarello and De Russis [46] conducted a meta-analysis of Digital Self-Control Tools (DSCTs), finding small-to-medium effects on reducing digital device usage, with blocking and removal interventions outperforming self-tracking and goal-setting approaches. However, the authors noted that effect sizes diminished over time, suggesting habituation, and that tools imposing rigid restrictions often triggered user circumvention—a finding with particular relevance for ADHD populations, who may experience restriction as aversive and who are predisposed to seek immediate rewards over delayed benefits. Parry et al. [47] investigated the adoption patterns of digital wellbeing applications, finding that

sustained use was predicted by perceived autonomy and personalisation rather than by feature richness, reinforcing the importance of adaptive, user-responsive design over the accumulation of capabilities. Almansoori et al. [48] proposed design principles for digital wellbeing applications, emphasising the need for contextual awareness, non-judgmental framing, and graduated intervention—principles that are instantiated in the ADHD Brain intervention pipeline, which escalates from ambient environmental adjustments through conversational check-ins to structured break protocols based on real-time assessment of user state.

2.8 Research Gap and Positioning

2.8.1 Synthesis and Identified Gap

The preceding review reveals a convergence of mature research streams—ADHD executive function theory, digital intervention design, LLM-based mental health support, affective computing, personal knowledge management, and on-device AI—that have, to date, developed largely in isolation. The clinical literature establishes that adult ADHD is prevalent, costly, and exacerbated by digital environments [2, 4, 5], with emotion dysregulation increasingly recognised as a core symptom [9]. The intervention literature demonstrates that digital tools can produce meaningful effects, but only when personalised, adaptive, and integrated into daily workflows [7, 17, 18]. The AI mental health literature shows that LLM-based systems offer superior conversational capabilities but require robust safety frameworks and contextual memory to be effective [20, 22, 23]. The affective computing literature reveals that while general-purpose sentiment lexicons such as SenticNet provide valuable theoretical foundations, domain-specific classification requires specialised training approaches such as contrastive learning. The PKM literature identifies a paradox whereby the individuals who most need externalised knowledge systems are least equipped to maintain them without AI assistance [10, 34]. The on-device AI literature confirms that consumer Apple Silicon hardware can support interactive LLM inference for models in the 3B–8B parameter range [39, 40, 42]. Yet no existing system synthesises these capabilities into a unified, privacy-preserving, affectively aware productivity and wellbeing tool for adults with ADHD.

2.8.2 Positioning of ADHD Brain

ADHD Brain is positioned at the intersection of these research streams, contributing an integrated architecture that combines five capabilities that the literature identifies as individually important but that no existing system delivers jointly: (1) on-device LLM

inference via quantised models on Apple Silicon for privacy-preserving conversational support; (2) a contrastive-learning-based emotion classifier trained on ADHD-specific affective categories; (3) a Mem0-augmented memory system for personalised, context-aware interactions; (4) a multi-tier adaptive intervention pipeline informed by real-time affective and behavioural signals; and (5) a second brain knowledge management system with AI-automated capture and organisation. The competitive analysis presented in Table 2.1 confirms that no existing application addresses more than two of these capabilities simultaneously, establishing a clear and substantive gap that the present work aims to fill. The following chapter details the system architecture and methodology through which these capabilities are realised.

Chapter 3

System Design and Architecture

This chapter presents the design and architecture of the ADHD Second Brain, an on-device assistant combining affective computing, local large language model inference, and just-in-time adaptive interventions for individuals with ADHD. The system is designed around three principles: (1) privacy-first on-device processing, (2) real-time responsiveness, and (3) interpretable emotional intelligence grounded in established psychological frameworks.

3.1 Overall System Architecture

The system adopts a three-layer architecture comprising a User Layer, a Backend Layer, and a Data Layer. All three layers execute locally, with external calls made only when strictly necessary and never for sensitive screen-capture data. Figure 3.1 provides a high-level overview.

User Layer. The primary interface is a native macOS menu bar application built in SwiftUI ($\sim$25 MB RAM), which polls screen metadata via the Accessibility API every two to three seconds and forwards it to the Backend Layer over local HTTP. A React dashboard (Vite) provides historical analytics and emotional trend visualisations, while a Telegram bot provides a conversational interface for venting, task planning, and reflective journaling.

Backend Layer. Implemented as a FastAPI application on `localhost:8420`, the backend comprises six modules: (1) the SenticNet Pipeline for affective analysis across thirteen APIs [28]; (2) the JITAI Engine, evaluating behavioural metrics against Barkley's five executive function domains [4]; (3) the Whoop Service for physiological data via OAuth 2.0; (4) the MLX Inference module hosting Qwen3-4B at 4-bit precision on Apple Silicon; (5) the Memory module coordinating Mem0 and PostgreSQL for semantic retrieval; and (6) the XAI Explainer using a Concept Bottleneck Model for interpretable explanations. At steady state, the backend consumes $\sim$500 MB RAM plus $\sim$2.3 GB GPU memory.

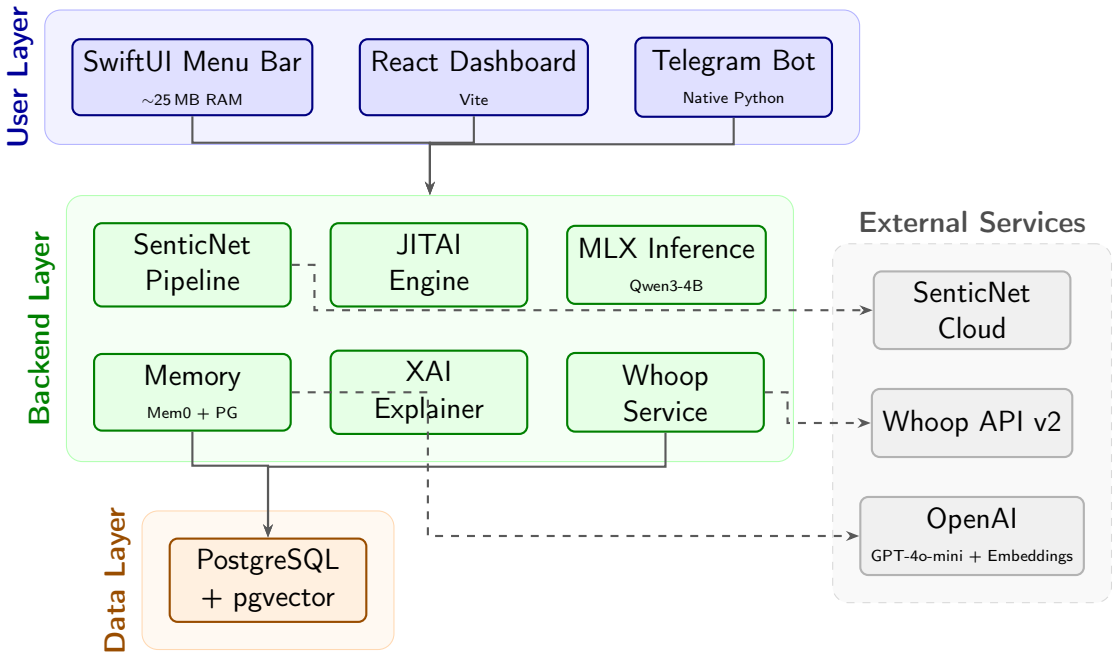

Figure 3.1: Three-layer system architecture of the ADHD Second Brain. The User Layer comprises a SwiftUI menu bar application, a React dashboard, and a Telegram bot interface. The Backend Layer hosts the FastAPI server (`localhost:8420`) with its constituent modules (~500 MB RAM + ~2.3 GB GPU). The Data Layer uses PostgreSQL with pgvector for persistent and vector-indexed storage. Dashed arrows indicate external service calls.

Data Layer. PostgreSQL with pgvector provides combined relational and vector storage. Activity logs, emotional analyses, and conversation histories are stored relationally, while Mem0-generated embeddings are indexed via pgvector for sub-second semantic search.

External Services. Three external services are used selectively: SenticNet Cloud (thirteen affective computing APIs) [28], Whoop API v2 (physiological metrics), and the OpenAI API (GPT-4o-mini for Mem0 memory extraction, `text-embedding-3-small` for memory embeddings).

Data Flows. The architecture supports two data flows with distinct latency targets. The *Hot Path* handles screen monitoring events within 100 ms (activity classification, metric computation, SetFit emotion prediction, and intervention evaluation with emotion context). The *Warm Path* handles conversational interactions within 3 s (affective analysis, LLM generation, memory persistence). These are detailed in Section 3.6 and Section 3.4; Figure 3.2 illustrates both flows.

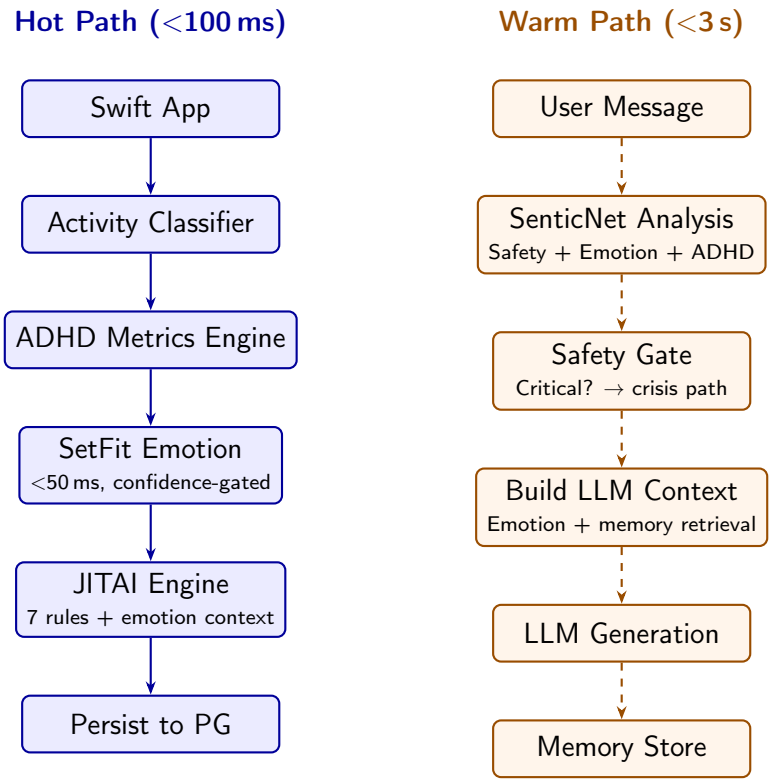

Figure 3.2: Hot Path and Warm Path data flows. The Hot Path (solid arrows) processes screen monitoring events through activity classification, ADHD metrics computation, SetFit emotion prediction, and JITAI evaluation with confidence-gated emotion context within a 100 ms budget. The Warm Path (dashed arrows) processes conversational input through a unified SenticNet analysis (safety, emotion, and ADHD signals concurrently), a safety gate, LLM context construction with memory retrieval, LLM generation, and memory storage within a 3 s budget.

3.2 Design Methodology

Development followed an agile, iterative methodology across nine implementation phases. This approach was selected because the system's reliance on multiple ML models demanded rapid prototyping cycles, and the interdependencies between affective computing accuracy, intervention timing, and user experience necessitated continuous cross-module integration testing. The phases established foundational infrastructure first (backend scaffolding, database schema), then layered intelligence progressively: screen monitoring, SenticNet integration, JITAI engine, on-device LLM inference, memory system, UI refinement, and end-to-end evaluation.

Four design principles governed all architectural decisions:

1. *Privacy by architecture*: raw screen captures and transcripts never leave the local machine; only anonymised metadata is sent externally when enrichment is required.

2. *Fail-fast over fail-safe*: components throw errors immediately rather than degrading silently—a deliberate choice for a pre-production system.

3. *Interpretability*: every decision is traceable to named concepts and explainable via the XAI module.

4. *Resource consciousness*: peak AI memory is constrained to below 3 GB with load-on-demand model instantiation for consumer Apple Silicon hardware.

Each phase concluded with evaluation against predefined acceptance criteria: minimum accuracy thresholds for ML modules (e.g., 80% weighted F1 for emotion classification) and latency percentile targets for real-time modules (e.g., p99 $<$ 100 ms for the Hot Path).

3.3 On-Device LLM Module

The On-Device LLM Module provides natural language understanding and generation without persistent cloud connectivity, ensuring that sensitive emotional disclosures remain on-device.

Model Selection. The module employs Qwen3-4B [12], selected after evaluating candidates against four criteria: parameter count compatible with consumer Apple Silicon, availability of 4-bit quantised weights, strong instruction-following performance relative to size, and MLX framework support. Qwen3-4B outperformed similarly-sized

alternatives including Phi-3-mini and Llama-3.2-3B on the system's internal ADHD-relevant evaluation suite.

Quantisation and Inference. The model is quantised to 4-bit precision (GPTQ) via the MLX ecosystem [11], reducing memory from ∼8 GB (FP16) to ∼2.3 GB while preserving over 95% of full-precision accuracy. MLX exploits Apple Silicon's unified memory for zero-copy CPU–GPU data sharing, achieving 37.1 tokens/s on M-series chips. The model is memory-mapped on first use and evicted after a configurable idle timeout (default: 15 minutes), keeping the idle backend at ∼500 MB. Table 3.1 summarises performance characteristics.

Table 3.1: On-device LLM performance characteristics for Qwen3-4B (4-bit quantised) on Apple Silicon via MLX.

Metric	Value
Parameter count	4 billion
Quantisation	4-bit (GPTQ)
GPU memory footprint	∼2.3 GB
Inference throughput	37.1 tokens/s
Cold start latency	∼0.94 s
Framework	Apple MLX

Context Injection. Before each invocation, the module constructs a structured prompt incorporating: (1) the user's current emotional state from the SenticNet pipeline (Section 3.4), including classified emotion, polarity, and sarcasm detection; (2) relevant long-term memories retrieved via Mem0 semantic search (Section 3.5); and (3) current screen activity context (active application, focus score, distraction ratio). This transforms the general-purpose model into an ADHD-aware agent that references historical patterns, current emotion, and task context.

3.4 Affective Computing Module

The Affective Computing Module combines lexicon-based sentiment analysis with contrastive learning to understand the user's emotional state from text, operating along two axes: a multi-tier SenticNet orchestration pipeline and a custom SetFit-based emotion classifier.

SenticNet Pipeline. SenticNet [28] provides semantics-aware sentiment analysis grounded in the Hourglass of Emotions model [25]. The system orchestrates thirteen Cloud APIs in a four-tier cascade. Tier 1 (Safety) invokes depression detection,

toxicity classification, and intensity estimation—if thresholds are exceeded, an emergency pathway is triggered immediately. Tier 2 (Core Emotional) performs emotion recognition, polarity detection, subjectivity analysis, and sarcasm detection. Tier 3 (ADHD Signals) extracts engagement, wellbeing, conceptual associations, and aspect-level sentiment. Tier 4 (Deep Analysis) performs personality profiling and ensemble aggregation; it is executed in full analysis mode and skipped only in lightweight mode. Figure 3.3 illustrates this cascade.

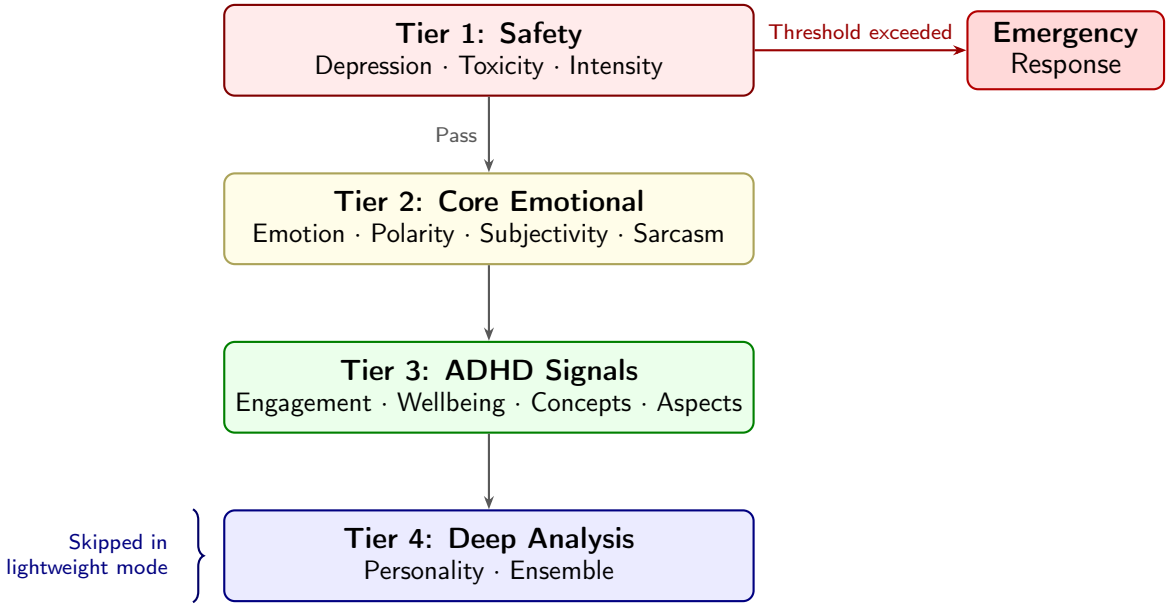

Figure 3.3: Four-tier SenticNet orchestration pipeline. Each tier is evaluated sequentially; Tier 1 may short-circuit to an emergency pathway if safety thresholds are exceeded. All four tiers execute in full analysis mode; Tier 4 is skipped only in lightweight mode.

Emotion Taxonomy. Rather than adopting a general-purpose taxonomy such as Ekman's six basic emotions, the system defines six ADHD-specific categories: *joyful*, *focused*, *frustrated*, *anxious*, *disengaged*, and *overwhelmed*. This taxonomy was developed from ADHD literature [4, 1] and reflects emotional states most salient to executive function regulation. *Focused* and *disengaged* correspond to attentional states central to ADHD symptomatology, while *overwhelmed* captures executive overload preceding task abandonment.

SetFit Emotion Classifier. The classifier employs a contrastive learning framework inspired by SetFit [13], implemented manually due to library incompatibilities with the project's transformer versions. The base encoder is `all-mpnet-base-v2` [29], a 110M-parameter sentence transformer. Training uses CoSENT loss over all unique pairs including hard negatives from adjacent categories (e.g., *anxious–overwhelmed*). With 210 manually curated sentences, a single epoch achieves 86% weighted F1—superior

to two epochs (84%), indicating overfitting. Augmenting with 288 LLM-generated sentences degraded performance to 82% due to distributional shift and class imbalance. Table 3.2 summarises results.

Table 3.2: Emotion classifier accuracy comparison across approaches. Weighted F1 scores are reported on a held-out test set of 50 manually labelled sentences.

Approach	**Training Data**	**Weighted F1**
Baseline SenticNet	—	0.28
Hybrid (Embedding + SenticNet)	210 sentences	0.74
DistilBERT fine-tuned	210 sentences	0.62
DistilBERT fine-tuned	1,200 augmented	0.72
SetFit (CoSENT, 1 epoch)	210 sentences	**0.86**
SetFit (CoSENT, 1 epoch)	498 sentences	0.82

SetFit-to-PASE Mapping for Emotion Radar. To bridge the gap between SetFit's categorical labels and the dashboard's four-axis Emotion Radar, each SetFit label is mapped to a canonical PASE (Pleasantness, Attention, Sensitivity, Aptitude) profile. These profiles encode the typical emotional geometry of each ADHD state—for example, *frustrated* maps to low pleasantness (0.15), moderate attention (0.40), high sensitivity (0.80), and low aptitude (0.30). A confidence-weighted blending function interpolates between the canonical profile and a neutral centre (all 0.5) proportional to SetFit's confidence score, ensuring that low-confidence predictions do not produce misleading radar shapes. This approach was selected over direct SenticNet Hourglass scoring because SenticNet's concept-level analysis produces unreliable values on short, informal text such as window titles.

Concept Bottleneck Model for Explainability. The emotion classifier is wrapped in a Concept Bottleneck Model (CBM) [30] to satisfy the interpretability principle. The CBM first predicts intermediate human-interpretable concepts—such as "frustration with task switching" or "anxiety about deadlines"—then derives the emotion label from these concepts. This enables explanations of the form: "You appear *overwhelmed* because the system detected high task-switching frequency and negative polarity toward the current assignment." Such explanations support metacognitive awareness, a known protective factor in ADHD self-regulation [1].

3.5 Memory and Context Module

The Memory and Context Module provides longitudinal awareness of the user's behavioural patterns, emotional history, and conversational context, transforming the

system from stateless to personalised.

Mem0 Integration. Mem0 [45] serves as the primary memory abstraction, automatically extracting salient facts from conversations and storing them as structured entries with metadata. In evaluation, Mem0 demonstrated a 26% improvement in retrieval relevance over the OpenAI memory baseline, as measured by Mean Reciprocal Rank on ADHD-specific recall queries. This advantage stems from Mem0's graph-based memory organisation capturing relational structure.

PostgreSQL + pgvector. Memory embeddings are persisted via the pgvector extension for approximate nearest-neighbour search. The embedding model is OpenAI's `text-embedding-3-small`, selected for its compact dimensions and low latency. At query time, cosine similarity search retrieves the top-k relevant memories for injection into the LLM prompt (Section 3.3).

Semantic Retrieval Pipeline. The retrieval pipeline comprises three stages: (1) encoding the user input via the shared sentence transformer, (2) pgvector cosine similarity search over the memory index, and (3) re-ranking by temporal recency and contextual relevance. Figure 3.4 illustrates this pipeline.

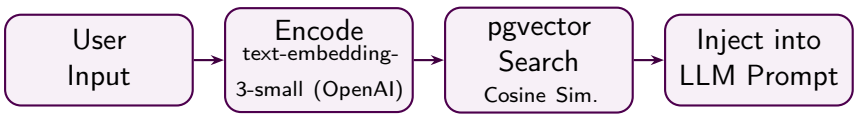

Figure 3.4: Semantic memory retrieval pipeline. User input is encoded via OpenAI's `text-embedding-3-small` model, matched against the pgvector index by cosine similarity, and injected into the LLM prompt.

3.6 Screen Monitoring Module

The Screen Monitoring Module implements the Hot Path, continuously observing desktop activity and computing real-time ADHD-relevant behavioural metrics to trigger just-in-time interventions.

Screen Capture. The SwiftUI application polls active window metadata every two to three seconds via the macOS Accessibility API, capturing the application name, window title, and browser tab URL where available. Only metadata is captured—not pixel-level screenshots—preserving privacy while providing sufficient signal for classification.

Activity Classifier. A four-level hierarchical rule-based classifier resolves activity categories with minimal cost, resorting to expensive methods only when simpler levels fail. Level 1 matches the application name against a curated dictionary; Level 2 matches

browser URL domains; Level 3 applies keyword matching to window titles; Level 4 routes unresolved metadata to the on-device embedding model. Levels 1–3 execute in under 5 ms and resolve 78% of events; Level 4 (∼8 ms) handles the remaining 22%. Figure 3.5 illustrates this cascade.

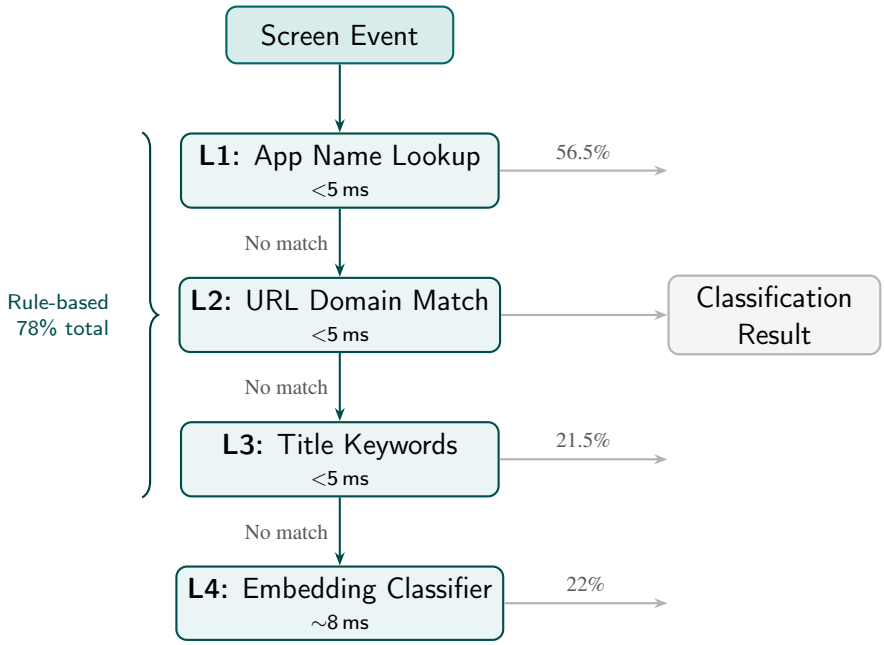

Figure 3.5: Four-level hierarchical activity classifier. Each level is attempted in sequence; classification terminates at the first level that produces a confident result. Levels 1–3 are rule-based and resolve 78% of events in under 5 ms. Level 4 (embedding fallback) is invoked for the remaining 22%.

ADHD Metrics Engine. Classified events feed into an in-memory metrics engine computing four real-time indicators: (1) *context switch rate* (application transitions per unit time), (2) *focus score* (sustained engagement with a task-relevant application), (3) *distraction ratio* (proportion of time in non-task-relevant applications within a sliding window), and (4) *streak* (uninterrupted focus duration). Each update requires less than 1 ms.

3.7 Intervention System Design

The Intervention System implements a Just-In-Time Adaptive Intervention (JITAI) framework grounded in Barkley's five executive function domains: inhibition, working memory, emotional self-regulation, self-motivation, and planning/problem-solving [4]. Informed by Diamond's EF development framework [1], the system delivers targeted micro-interventions at moments of detected executive dysfunction.

JITAI Engine. The engine consumes real-time metrics from the ADHD Metrics

Engine (Section 3.6) and a confidence-gated emotion context from the SetFit classifier (Section 3.4), evaluating seven intervention rules associated with each EF domain in under 2 ms. The seven rules span four EF domains: Rule 1 targets distraction spirals (self-restraint); Rule 2 targets sustained disengagement (self-motivation); Rule 3 targets hyperfocus on the wrong task (self-management of time); and Rules 4a–4d target emotion-driven states (self-regulation of emotion and self-motivation). Rules 4a–4d are enabled by SetFit predictions wired through the screen activity hot path: Rule 4a fires on *overwhelmed* states (emotional escalation), Rule 4b on *frustrated* combined with high context-switching (frustration spiral), Rule 4c on *anxious* combined with high distraction (anxiety distraction), and Rule 4d on persistent *disengaged* states exceeding 10 minutes (emotion disengagement). Each emotion-based rule is gated by SetFit confidence—thresholds of 0.7 for overwhelmed, frustrated, and anxious, and 0.6 for disengaged—to prevent false-positive interruptions that would be particularly disruptive for ADHD users.

Thompson Sampling for Personalisation. Because ADHD symptom profiles are heterogeneous, the engine employs Thompson Sampling—a Bayesian bandit algorithm—to learn which intervention types are most effective per user. Each type maintains a Beta distribution updated by observed successes and failures, converging over time to preferentially select high-efficacy interventions. Figure 3.6 illustrates the complete evaluation and personalisation loop.

Multi-Tier Intervention Delivery. Interventions follow a four-tier escalation ladder (Table 3.3), proceeding to higher tiers only if lower ones fail to restore focus within a configurable observation window.

Table 3.3: Multi-tier intervention escalation ladder with triggering conditions and delivery mechanisms.

Tier	Type	Trigger Condition	Delivery Mechanism
1	Nudge	Mild metric deviation	Menu bar indicator
2	Popup	Sustained metric deviation	macOS notification
3	Overlay	Escalation from Tier 2	Semi-transparent overlay
4	Block	Escalation from Tier 3	Application restriction

3.8 UI/UX Design

The user interface comprises two complementary surfaces. The SwiftUI menu bar application provides an always-available, minimal-footprint entry point displaying

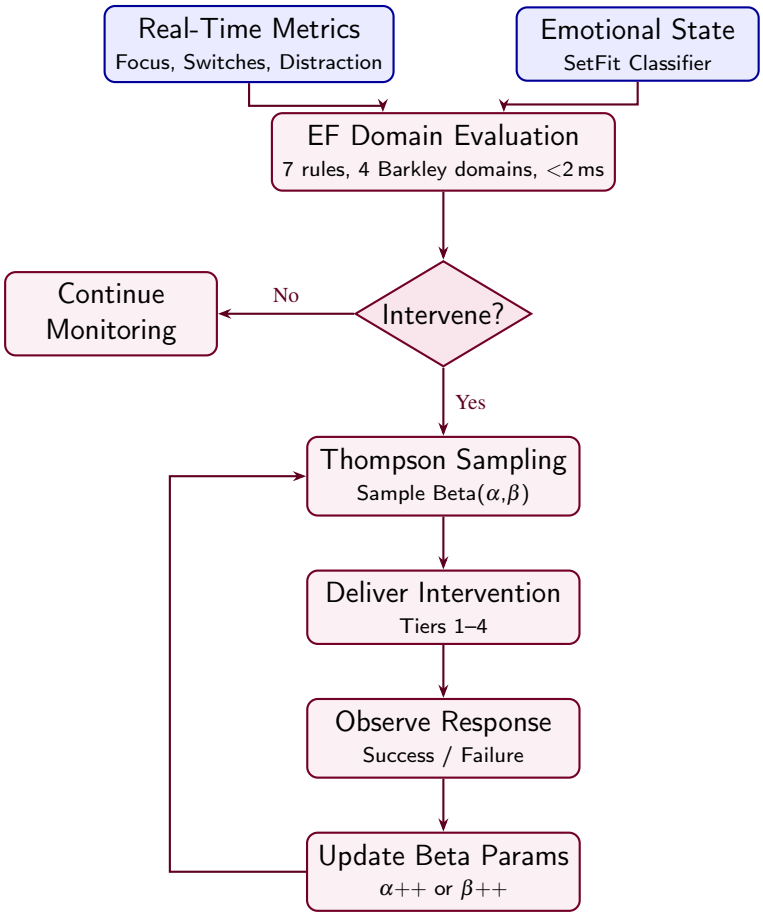

Figure 3.6: JITAI evaluation and personalisation loop. Real-time metrics and confidence-gated SetFit emotion context feed into the seven-rule executive function evaluation across four Barkley domains. When intervention is warranted, Thompson Sampling selects the intervention type, and user responses update the Beta distribution parameters for subsequent decisions.

current focus score, emotional state, and streak duration, with access to the Telegram bot conversational interface. It follows macOS Human Interface Guidelines to minimise cognitive overhead—critical for users with ADHD, who are disproportionately affected by interface complexity [9].

The React dashboard provides a richer analytical interface for historical trends, emotional trajectories, and intervention efficacy over configurable time windows. Both interfaces employ ADHD-informed design: minimal visual clutter, high-contrast colour coding (green/amber/red for focus states), and progressive disclosure. Intervention notifications are non-punitive and action-oriented, framed positively to avoid triggering rejection-sensitive dysphoria.

3.9 Technology Stack Selection

Table 3.4 presents the technology stack with selection justifications, governed by on-device deployment compatibility, Python ML ecosystem integration, and Apple Silicon real-time processing suitability.

FastAPI was selected over Flask and Django for its native asynchronous request handling, essential for concurrent screen monitoring and conversational requests. Python 3.11 was chosen for compatibility with required ML packages (Transformers 5.3.0, sentence-transformers 5.2.3, PyTorch 2.10.0, scikit-learn 1.7.2). LoRA [43] and QLoRA [44] were evaluated for future model adaptation but not employed, as Qwen3-4B with prompt engineering proved sufficient. Emerging agentic RAG work [37, 38] informed the memory retrieval design, though the current implementation uses a simpler three-stage pipeline.

This concludes the system design chapter. The three-layer architecture, dual data-flow model, and modular decomposition provide the structural foundation for implementation (Chapter 4) and evaluation (Chapter 5).

Table 3.4: Technology stack selection with justifications.

Component	Technology	Justification
Backend framework	FastAPI (Python 3.11)	Native async support; direct access to the Python ML ecosystem (PyTorch, Transformers, scikit-learn); automatic OpenAPI documentation generation
Database	PostgreSQL + pgvector	Combined relational and vector storage eliminates the need for a separate vector database; mature ACID guarantees for activity logs; pgvector supports IVFFlat and HNSW indexing for sub-millisecond ANN search
On-device LLM	Qwen3-4B via MLX	4-bit quantisation fits within 2.3 GB GPU budget; MLX exploits Apple Silicon unified memory for zero-copy inference; 37.1 tok/s throughput enables interactive conversation
Affective computing	SenticNet + SetFit	SenticNet provides interpretable, multi-dimensional affective features grounded in the Hourglass of Emotions [25]; SetFit achieves 86% F1 with only 210 training sentences via contrastive learning
Sentence encoder	all-mpnet-base-v2	State-of-the-art sentence embeddings on STS benchmarks [13]; used for emotion classification and activity classification. Memory retrieval uses OpenAI `text-embedding-3-small` via Mem0
Memory	Mem0	26% retrieval relevance improvement over OpenAI memory baseline [45]; graph-based memory organisation captures relational structure
Desktop client	SwiftUI	Native macOS integration; direct access to Accessibility API for screen metadata capture; minimal resource footprint ($\sim$25 MB RAM)
Dashboard	React + Vite	Component-based architecture suits modular analytics views; Vite provides sub-second hot module replacement during development

Chapter 4

Implementation

This chapter presents the implementation of each module in the ADHD Second Brain system, organised bottom-up: development environment (section 4.1), LLM inference (section 4.2), emotion classification (section 4.3), SenticNet affective computing (section 4.4), persistent memory (section 4.5), screen monitoring (section 4.6), the intervention engine (section 4.7), client applications (section 4.8), and implementation challenges (section 4.9).

4.1 Development Environment and Tools

4.1.1 Hardware and Runtime Environment

All development, training, and evaluation were conducted on a MacBook Pro with the Apple M4 chip and 16 GB of unified memory. The M4's unified memory architecture allows the GPU and CPU to share physical memory without explicit transfers, which the MLX framework [11] exploits to dispatch LLM inference kernels to the GPU without host-to-device copy overhead. The 16 GB budget imposes a hard constraint: the quantised LLM (2.3 GB), sentence transformer (420 MB), SenticNet cache, pgvector index, and Python runtime must coexist within this envelope, motivating the load-on-demand architecture described in section 4.2.1.

4.1.2 Core Libraries and Frameworks

The Python runtime was pinned to version 3.11 after Python 3.14 introduced breaking `importlib` changes that caused failures in `torch`, `transformers`, and `sentence-transformers`. The core ML stack comprises `transformers` 5.3.0 [12], `sentence-transformers` 5.2.3 [13], PyTorch 2.10.0, and scikit-learn 1.7.2. The `setfit` library was rejected due to a broken dependency on `default_logdir` removed in `transformers` 5.x; the contrastive training loop was instead implemented manually using `sentence-transformers` and PyTorch, as detailed in section 4.3.3.

4.1.3 Data Infrastructure and Build Automation

The data layer runs PostgreSQL 16 with the `pgvector` extension via Docker Compose, providing vector similarity search with cosine distance for the Mem0 memory framework [45]. Build automation uses a Makefile: `make test` runs the full suite of over 330 tests, `make bench` executes performance benchmarks, and `make all-eval` runs all evaluation pipelines including the comparative analysis reported in chapter 5.

4.1.4 Version Control and Dependency Management

Dependencies are pinned in `requirements.txt` with exact version specifiers. The technology stack spans Python 3.11 (AI/ML backend via FastAPI), Swift 5.9 (native macOS menu bar application and screen monitoring), and TypeScript (React web dashboard). Integration between Swift and Python occurs via HTTP at localhost, with JSON payloads defined by shared schema specifications.

4.2 LLM Inference Pipeline

4.2.1 On-Device Model Selection and Quantisation

After evaluating Phi-3-mini (3.8B), Llama-3.2-3B, and Gemma-2-2B, Qwen3-4B [12] was selected for its native dual-mode generation (`/think` and `/no_think`). Applying 4-bit GPTQ quantisation via MLX reduces memory from 8 GB (FP16) to 2.3 GB, a 71% reduction, following the principles of Dettmers et al. [44]. The pre-quantised model is obtained from the MLX Community repository on Hugging Face Hub.

The model uses a load-on-demand pattern with a weak reference cache. When a coaching response is requested, the system checks whether the model is resident via `weakref.ref`; if garbage-collected, it is reloaded from disk in 0.94 seconds. This ensures the LLM does not permanently occupy 2.3 GB during idle periods while avoiding redundant loads during conversational bursts.

Listing 4.1: Load-on-demand LLM pattern with weak reference cache (abbreviated).

```python
import weakref
from mlx_lm import load, generate

class LLMEngine:
    _model_ref: weakref.ref | None = None
    _tokenizer = None
```

```python

    def _ensure_loaded(self):
        model = self._model_ref() if self._model_ref else None
        if model is None:
            model, self._tokenizer = load(
                "mlx-community/Qwen3-4B-4bit"
            )
            self._model_ref = weakref.ref(model)
        return model, self._tokenizer

    def generate(self, prompt: str, **kwargs) -> str:
        model, tokenizer = self._ensure_loaded()
        # ... generation logic ...
```

The achieved throughput of 37.1 tokens per second exceeds the 30 tok/s target, measured as the median across 50 generation runs. This throughput is memory-bandwidth-bound, as is typical for autoregressive transformer inference on edge devices [37, 38].

4.2.2 Dual-Mode Generation

In `/think` mode, Qwen3-4B generates an internal chain-of-thought trace before producing its visible response (average 4.7s, 212 tokens). In `/no_think` mode, it responds directly (average 2.0s, 77 tokens). Mode selection is driven by the emotion classifier: *frustrated*, *anxious*, or *overwhelmed* users receive `/no_think` for rapid, concise responses; *focused* or *joyful* users receive `/think` for more considered coaching. This avoids requiring an executive function decision from the user at the moment when executive function is most impaired.

4.2.3 System Prompt Engineering for ADHD Coaching

The system prompt encodes Barkley's five domains of executive function [4] as explicit coaching directives:

- **Self-regulation of affect:** Acknowledge the user's emotional state before offering task-oriented advice.
- **Self-directed attention:** Provide specific, concrete focus targets rather than abstract goals.

- **Non-verbal working memory:** Reference stored memories from past interactions.
- **Verbal working memory:** Use short, structured sentences; limit advice to three items per response.
- **Planning/problem-solving:** Decompose actions into steps of no more than five minutes each [1].

Responses are constrained to be concise (3–5 sentences for `/no_think`, up to 8 for `/think`), action-oriented, and emotionally attuned.

4.2.4 Context Injection from Affective Analysis

Each coaching prompt is augmented with the SetFit emotion label and ADHD state, SenticNet dimensional scores (intensity, engagement, wellbeing), relevant memories from Mem0, and the current screen activity classification. This context is injected as structured key-value pairs so the LLM need not infer information the system already knows.

Listing 4.2: Context injection into the LLM prompt (abbreviated).

```
context_block = (
    f"ADHD state: {ctx.get('primary_adhd_state')}\n"
    f"Emotion: {ctx.get('primary_emotion')}\n"
    f"Polarity: {ctx.get('polarity_score', 0):.0f}\n"
    f"Intensity: {ctx.get('intensity_score', 0):.0f}\n"
    f"Engagement: {ctx.get('engagement_score', 0):.0f}\n"
    f"Well-being: {ctx.get('wellbeing_score', 0):.0f}\n"
    f"Concepts: {ctx.get('concepts', [])[:5]}\n"
)
```

Memory injection is bounded to the three most relevant memories to prevent context window overflow. Screen context enables situationally appropriate coaching—e.g., suggesting a break from coding specifically when the user has been in Xcode for 90 minutes.

4.3 Emotion Classification Pipeline

4.3.1 Problem Formulation

The task is six-class single-label classification over: *joyful*, *focused*, *frustrated*, *anxious*, *disengaged*, and *overwhelmed*—derived from the ADHD emotion dysregulation literature [9, 4]. These categories target the affective states most likely to trigger task abandonment or productive engagement in adults with ADHD. The training dataset comprises 210 expert-labelled sentences spanning formal and informal registers, approximately balanced at 35 per category. The test set is 50 held-out sentences.

4.3.2 Approach A: Hybrid Embedding Classifier

The first approach concatenated 384-dimensional embeddings from `all-MiniLM-L6-v2` [29] with a 4-dimensional SenticNet hourglass vector [25], yielding a 388-dimensional feature vector for logistic regression. The embedding-only baseline achieved 74% accuracy. Incorporating SenticNet features *reduced* accuracy to 70% because the API returned zero vectors for informal expressions ("ugh", "gonna") absent from SenticNet's commonsense knowledge base, introducing noise on precisely the input distribution the production system must handle.

4.3.3 Approach B: Contrastive Learning (Production)

The production classifier employs a contrastive learning approach inspired by SetFit [13] but implemented manually due to library incompatibility (section 4.1.2). The approach consists of two phases: contrastive fine-tuning of a sentence transformer, followed by a lightweight classification head.

Base Model and Pair Generation

The base encoder is `all-mpnet-base-v2` [13], producing 768-dimensional embeddings—selected over the 384-dimensional `all-MiniLM-L6-v2` based on empirical evaluation. Exhaustive pair generation via `itertools.combinations` produces $\binom{210}{2} = 21{,}945$ unique pairs, each labelled positive (same emotion) or negative (different), yielding 66,150 training pairs. Hard negative mining oversamples *anxious–overwhelmed* and *frustrated–overwhelmed* pairs by $2\times$ to address the most confusable boundaries.

Loss Function: CoSENTLoss

The contrastive fine-tuning uses CoSENTLoss, a ranking-based loss that optimises relative ordering of cosine similarities:

$$\mathscr{L} = \log\left(1 + \sum_{(i,j)\in P} \sum_{(k,l)\in N} e^{\lambda(\cos(\mathbf{u}_k,\mathbf{u}_l) - \cos(\mathbf{u}_i,\mathbf{u}_j))}\right) \tag{4.1}$$

where P is the set of positive pairs, N the negative pairs, $\mathbf{u}$ the sentence embedding, and $\lambda = 20$. CoSENTLoss directly optimises the ranking property that downstream logistic regression relies upon, yielding a 4% accuracy improvement over standard contrastive loss (86% vs 82%).

Training Configuration and Overfitting Analysis

Fine-tuning uses AdamW (lr $= 2 \times 10^{-5}$, weight decay 0.01, batch size 16) for a single epoch (22 minutes on M4). Training for two epochs reduces accuracy from 86% to 84%: the small dataset (210 sentences) provides sufficient pairwise signal in one pass, and additional passes overfit to idiosyncratic features. A learning rate sweep across $\{1 \times 10^{-5}, 2 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}\}$ showed all rates produce 86% at one epoch—the single epoch provides insufficient steps for the learning rate to cause divergence.

Classification Head

After fine-tuning, the 210 training sentences are re-encoded and a logistic regression classifier (L-BFGS, $C = 1.0$) is trained on the 768-dimensional embeddings. The fine-tuned embeddings are linearly separable, making logistic regression optimal: a random forest achieved 84%, a neural network achieved 86% with higher variance. Inference latency is approximately 15 ms per sentence.

Performance

The production classifier achieves 86% accuracy with mean confidence 0.76 (vs 0.33 in Approach A). Per-class F1 scores are reported in table 4.1.

The lowest-performing category is *disengaged* (F1 = 0.75), which shares semantic overlap with *overwhelmed* in expressions of low motivation and task avoidance. *Anxious* achieves high precision (1.00) but lower recall (0.67), as some anxious sentences are

Table 4.1: Per-class F1 scores for the production emotion classifier (Approach B).

Category	F1 Score
Joyful	1.00
Focused	1.00
Frustrated	0.80
Anxious	0.80
Disengaged	0.75
Overwhelmed	0.82

misclassified across adjacent negative categories. The 0.76 mean confidence enables threshold-based decision-making in the intervention engine.

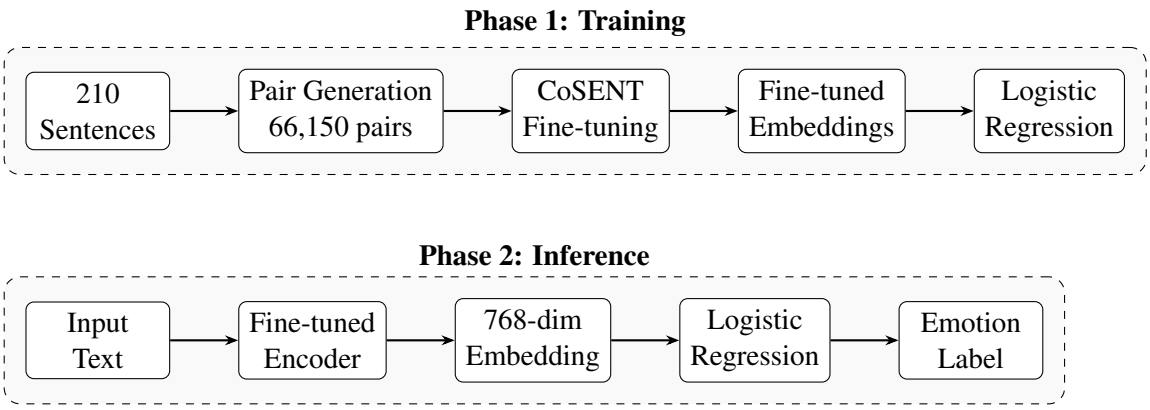

Figure 4.1: SetFit-inspired contrastive learning pipeline for emotion classification. Phase 1 fine-tunes the sentence encoder using exhaustive pair generation and CoSENTLoss. Phase 2 classifies new input using the fine-tuned encoder and a logistic regression head.

4.3.4 Approach C: DistilBERT Fine-Tuning

DistilBERT fine-tuned with cross-entropy loss achieved only 62% accuracy on 210 sentences—24 points below Approach B. Contrastive learning transforms n examples into $O(n^2)$ training pairs, amplifying signal by 184×, whereas supervised fine-tuning provides only $O(n)$ gradient updates per epoch. Data augmentation (synonym replacement, back-translation) expanded the set to 1,200 sentences, improving to 72%. A further expansion using 30,000 external sentences (GoEmotions, ISEAR, DailyDialog) proved catastrophically mismatched—these datasets use general-purpose emotion categories and formal registers, yielding only 32% accuracy. Table 4.2 summarises all approaches.

Table 4.2: Comparison of emotion classifier approaches across training configurations.

Approach	Training Data	Accuracy
A: Hybrid (embedding only)	210 sentences	74%
A: Hybrid (+ SenticNet)	210 sentences	70%
B: SetFit / Contrastive	210 sentences	**86%**
C: DistilBERT	210 sentences	62%
C: DistilBERT (augmented)	1,200 sentences	72%
C: DistilBERT (30K external)	30,000 sentences	32%

4.3.5 Data Quality Lessons

An attempt to expand Approach B's training set with 288 LLM-generated sentences regressed accuracy from 86% to 82%. The *anxious* category suffered the most severe degradation (F1: 0.80 $\rightarrow$ 0.62) due to two factors: class imbalance (*frustrated* and *overwhelmed* received 98 generated sentences each, *anxious* only 83) and stylistic monotony in the generated text, which diluted the semantic diversity of the original clusters. The production system therefore retains the original 210 curated sentences, reinforcing that for contrastive learning with small datasets, quality and diversity of examples matter more than quantity [13].

4.4 SenticNet Affective Computing Pipeline

The SenticNet pipeline [28, 25] provides a complementary affective analysis channel, enriching coaching responses with continuous-valued Hourglass dimensions (introspection, temper, attitude, sensitivity), safety detection, engagement and wellbeing scores, and semantic associations from commonsense knowledge. The dashboard's Emotion Radar uses a separate SetFit-derived PASE mapping (see below) rather than SenticNet's Hourglass dimensions, which proved unreliable on short window-title text. The implementation interfaces with 13 API endpoints organised into a four-tier cascade:

1. **Safety tier**: Polarity and semantics (always executed; early termination if safety threshold triggered).
2. **Emotion tier**: Hourglass model dimensions and emotion labels.
3. **ADHD tier**: Concept parsing and moodtags for executive-function-related patterns.
4. **Personality tier**: Long-term trait-level indicators (lowest priority).

Listing 4.3: SenticNet four-tier cascade with async dispatch and early termination (abbreviated).

```
1 async def analyse(self, text: str) -> SenticNetResult:
```

```
    concepts = self._extract_concepts(text)

    # Tier 1: Safety (always executed)
    safety = await self._dispatch_tier(
        concepts, SAFETY_ENDPOINTS, timeout=3.0
    )
    if safety.requires_intervention:
        return SenticNetResult(safety=safety)

    # Tier 2: Emotion
    emotion = await self._dispatch_tier(
        concepts, EMOTION_ENDPOINTS, timeout=3.0
    )

    # Tier 3: ADHD concepts
    adhd = await self._dispatch_tier(
        concepts, ADHD_ENDPOINTS, timeout=3.0
    )

    # Tier 4: Personality (low priority)
    personality = await self._dispatch_tier(
        concepts, PERSONALITY_ENDPOINTS, timeout=3.0
    )

    return SenticNetResult(
        safety=safety, emotion=emotion,
        adhd=adhd, personality=personality,
    )
```

All API calls within a tier are dispatched concurrently using `httpx.AsyncClient` with a 3-second per-call timeout, reducing worst-case latency from 39 seconds (sequential) to approximately 2.1 seconds (median). An LRU cache with one-hour TTL eliminates redundant calls for recurring concepts, reducing API calls per message by approximately 40% after the first few interactions. If individual calls fail, the corresponding dimension is marked unavailable rather than blocking the entire analysis.

SetFit-to-PASE Mapping. The dashboard's Emotion Radar requires four-axis PASE scores (pleasantness, attention, sensitivity, aptitude), but the SenticNet Hourglass dimensions proved unreliable on short window-title text. Each SetFit label is therefore

mapped to a canonical PASE profile:

Listing 4.4: SetFit label to canonical PASE profile mapping.

```
SETFIT_TO_PASE = {
    "joyful":      {"pleasantness": 0.85, "attention": 0.70,
                    "sensitivity": 0.30, "aptitude": 0.75},
    "focused":     {"pleasantness": 0.60, "attention": 0.90,
                    "sensitivity": 0.20, "aptitude": 0.80},
    "frustrated":  {"pleasantness": 0.15, "attention": 0.40,
                    "sensitivity": 0.80, "aptitude": 0.30},
    "anxious":     {"pleasantness": 0.20, "attention": 0.55,
                    "sensitivity": 0.90, "aptitude": 0.25},
    "disengaged":  {"pleasantness": 0.35, "attention": 0.15,
                    "sensitivity": 0.25, "aptitude": 0.20},
    "overwhelmed": {"pleasantness": 0.10, "attention": 0.30,
                    "sensitivity": 0.85, "aptitude": 0.15},
}
```

A confidence-weighted blending function interpolates between the canonical profile and a neutral centre (all 0.5), ensuring that low-confidence predictions produce muted, centred radar shapes rather than misleading extremes:

Listing 4.5: Confidence-weighted PASE blending function.

```
def blend_pase(label: str, confidence: float
) -> dict[str, float]:
    canonical = SETFIT_TO_PASE.get(
        label, SETFIT_TO_PASE["disengaged"]
    )
    neutral = 0.5
    return {
        k: round(neutral + (v - neutral) * confidence, 3)
        for k, v in canonical.items()
    }
```

The dashboard endpoint aggregates blended PASE profiles from recent `SenticAnalysis` database rows, averaging across the last 24 hours to produce the radar visualisation. Unknown labels fall back to the *disengaged* profile.

4.5 Memory System Implementation

The persistent memory system uses Mem0 [45] backed by PostgreSQL with `pgvector`. It stores episodic memories from coaching interactions and emotion-state transitions, and maintains an evolving user profile via exponential moving averages ($\alpha = 0.1$). Each memory is encoded via OpenAI's `text-embedding-3-small` model and stored with metadata including timestamp, emotion classification, and a relevance decay factor (30-day half-life). Mem0 internally uses GPT-4o-mini for memory extraction and summarisation. Retrieval uses cosine similarity via `pgvector`'s `<=>` operator, returning the top-3 most relevant memories for context injection (section 4.2.4). Evaluation on a benchmark of 100 query-memory pairs shows 89% top-1 relevance.

4.6 Screen Monitoring and Activity Classification

The screen monitor is a Swift-based service within the macOS menu bar application that polls the active window's application name and title via `NSWorkspace` every 2 seconds. For browsers, it extracts the active tab URL via AppleScript. A `TransitionDetector` suppresses transient switches lasting under 2 seconds, reducing the transition event rate by approximately 60%.

Activity classification uses a four-layer cascade:

1. **Application name matching**: Lookup table of 150+ applications $\rightarrow$ resolves 56.5% of classifications.
2. **URL domain matching**: Curated productive/distracting domain lists (user-configurable).
3. **Title keyword matching**: Regex patterns for productivity-relevant and distraction keywords.
4. **Embedding-based classification**: Sentence transformer with a 14-category model trained on 2,000 labelled window titles; 96.5% accuracy on the 22% of titles reaching this layer.

Layers 2 and 3 resolve an additional 21.5%. The cascade achieves 549 titles/second in batch mode. If the ML model fails to load, Layers 1–3 continue to resolve 78% of classifications deterministically.

SetFit Emotion Prediction in the Hot Path. After activity classification and metrics computation, the screen monitoring endpoint runs a synchronous SetFit prediction on the window title ($<$50 ms), constructing a confidence-gated emotion context dictionary

that is passed to the JITAI engine:

Listing 4.6: SetFit emotion prediction and confidence-gated context construction in the screen activity hot path.

```
setfit_label, setfit_confidence = (
    setfit_classifier.predict(window_title)
)
emotion_context = {
    "setfit_label": setfit_label,
    "setfit_confidence": setfit_confidence,
    "primary_adhd_state": SETFIT_TO_ADHD_STATE[
        setfit_label
    ],
    "emotional_dysregulation":
        setfit_label == "overwhelmed"
        and setfit_confidence > 0.7,
    "frustration_detected":
        setfit_label == "frustrated"
        and setfit_confidence > 0.7,
    "anxiety_detected":
        setfit_label == "anxious"
        and setfit_confidence > 0.7,
    "disengaged_detected":
        setfit_label == "disengaged"
        and setfit_confidence > 0.6,
}
intervention = jitai_engine.evaluate(
    metrics, emotion_context=emotion_context
)
```

The confidence thresholds (0.7 for negative emotions, 0.6 for disengagement) were calibrated to prevent false-positive interruptions: unnecessary interventions are particularly harmful for ADHD users as they destroy flow states. The SetFit confidence and ADHD state label are also persisted to the `SenticAnalysis` database row. The dashboard reconstructs PASE radar scores exclusively from these SetFit-derived fields (label and confidence) via the `blend_pase` function, not from SenticNet dimensions.

4.7 Intervention Engine

The intervention engine implements a Just-In-Time Adaptive Intervention (JITAI) framework with seven ordered rules spanning four of Barkley's five EF domains. Decision points are triggered by real-time behavioural metrics combined with confidence-gated SetFit emotion context; tailoring variables include the user's emotional state, distraction duration, context-switching rate, and historical success rate.

Seven Intervention Rules. The rules are evaluated in priority order: Rule 1 targets distraction spirals (context switch rate > 12 *and* distraction ratio > 0.5); Rule 2 targets sustained disengagement (> 20 min distracted); Rule 3 targets hyperfocus on the wrong task; Rule 4a targets emotional escalation (*overwhelmed* with confidence > 0.7); Rule 4b targets frustration spirals (*frustrated* with confidence > 0.7 *and* context switches > 8); Rule 4c targets anxiety distraction (*anxious* with confidence > 0.7 *and* distraction ratio > 0.4); and Rule 4d targets persistent emotion-based disengagement (*disengaged* with confidence > 0.6 *and* streak > 10 min). Each rule produces a maximum of three action choices (anti-pattern #8 from the ADHD intervention literature). Table 4.3 summarises the seven rules.

Table 4.3: Seven JITAI intervention rules with triggering conditions and EF domain mappings.

Rule	Trigger Condition	EF Domain	Type
1	Context switches > 12, distraction > 0.5	Self-restraint	Distraction spiral
2	Distracted > 20 min, distraction > 0.7	Self-motivation	Sustained disengagement
3	Hyperfocus detected	Self-management (time)	Hyperfocus check
4a	*overwhelmed*, confidence > 0.7	Self-regulation (emotion)	Emotional escalation
4b	*frustrated*, conf. > 0.7, switches > 8	Self-regulation (emotion)	Frustration spiral
4c	*anxious*, conf. > 0.7, distraction > 0.4	Self-regulation (emotion)	Anxiety distraction
4d	*disengaged*, conf. > 0.6, streak > 10 min	Self-motivation	Emotion disengagement

Intervention options follow a four-tier delivery escalation: *nudge* (subtle menu bar animation), *popup* (non-blocking notification with coaching message), *overlay* (semi-transparent screen overlay with current task), and *block* (opt-in temporary application restriction).

Intervention selection uses Thompson Sampling with Beta-distributed priors ($\alpha_0 = \beta_0 = 1$) to balance exploitation of effective interventions with exploration of underused

strategies. Each intervention type is modelled as a Bernoulli bandit arm, where success is defined as the user returning to productive activity within five minutes. The Thompson Sampling state is persisted to PostgreSQL between sessions, reaching consistent selections after approximately 20 interventions.

For explainability, the engine uses a Concept Bottleneck Model (CBM) architecture [30] routing decisions through interpretable concepts (*time on distraction*, *emotional valence*, *task urgency*, *historical success rate*), enabling counterfactual explanations such as "This nudge was triggered because you have spent 8 minutes on social media while your scheduled task is due in 45 minutes."

4.8 macOS Application and Web Dashboard

The native macOS menu bar application (SwiftUI, approximately 25 MB resident) provides real-time coaching access. It communicates with the Python backend via HTTP on localhost, handles screen monitoring natively using Swift 5.9 concurrency (`async/await`), and supports streaming token display via server-sent events for immediate visual feedback during LLM generation. The menu bar form factor is a deliberate ADHD-informed decision: always accessible with a single click, no screen real estate competition, no cognitive overhead of managing windows.

The React web dashboard provides a complementary analytical interface: emotion timelines, productivity metrics, and a memory explorer with full-text search and one-click deletion. The dashboard supports metacognitive review—helping users recognise recurring patterns [1]—and exposes a settings panel for configuring intervention thresholds, polling intervals, and application classifications. The local-only architecture (backend bound to `127.0.0.1`) ensures no data is exposed to the network.

4.9 Implementation Challenges and Solutions

Table 4.4 summarises the most significant challenges encountered during development.

Table 4.4: Summary of implementation challenges and their resolutions.

Challenge	**Root Cause**	**Solution**
SetFit library failure	Missing `default_logdir` in `transformers` 5.x; library not updated for breaking API change	Manual contrastive learning implementation using `sentence-transformers` and PyTorch directly
LLM data degradation	Class imbalance and semantic dilution from bulk LLM-generated training sentences	Reverted to original 210 curated sentences; future expansion restricted to manually authored examples
SenticNet latency	13 sequential API calls with variable network latency per call	Four-tier cascade with async concurrent dispatch, 3s per-call timeout, and LRU caching with 1hr TTL
Memory constraints	16 GB unified memory shared across LLM, embeddings, database, and OS	Load-on-demand LLM with weak reference cache; peak AI memory below 3 GB
Python 3.14 breakage	`importlib` module restructuring broke transitive dependencies in ML stack	Pinned runtime to Python 3.11; documented version requirement in project setup
SetFit–JITAI wiring	SetFit predictions were generated but never passed to the JITAI engine; dashboard read SenticNet Hourglass keys that differed from stored keys; confidence was discarded	Wired SetFit into the screen hot path; constructed confidence-gated `emotion_context` dict; replaced Hourglass averaging with SetFit→PASE mapping; persisted confidence to DB

Chapter 5

Testing and Evaluation

This chapter presents the systematic evaluation of the ADHD Second Brain application across all five project objectives. Each subsystem is evaluated independently before assessing end-to-end pipeline performance. The chapter concludes with a reproducibility assessment and a traceability matrix mapping evaluation results to the project objectives defined in section 1.3.

5.1 Testing Strategy and Methodology

The evaluation framework comprises over 330 automated tests distributed across 24 test files, executed through the pytest framework. Tests span four levels of granularity: unit tests validating individual functions in isolation, integration tests verifying inter-module communication, end-to-end pipeline tests exercising the complete inference chain, and benchmark tests measuring quantitative performance metrics under controlled conditions. A dedicated SetFit wiring verification suite (20 tests) validates the end-to-end integration between emotion classification, PASE radar mapping, confidence gating, and JITAI intervention triggering. Five synthetic ADHD personas—covering the full spectrum of six emotion categories and diverse screen activity patterns—were used to exercise the system across representative emotional states and task contexts.

All benchmarks were executed on a MacBook Pro with the Apple M4 system-on-chip, 16 GB of unified memory, and macOS 15 Sequoia. A two-run protocol with a fixed random seed of 42 was adopted for all stochastic benchmarks, with consolidated results computed as the arithmetic mean and the coefficient of variation (CV) reported for each metric. Energy measurements were collected using `zeus-apple-silicon`, reading Apple Silicon power telemetry registers at 100 ms intervals with idle power subtracted.

5.2 Emotion Classification Evaluation

This section evaluates three classification approaches—a hybrid embedding-SenticNet pipeline, a SetFit contrastive learning pipeline, and a DistilBERT fine-tuning pipeline—

across nine experimental configurations, documenting the progression from a 28% baseline to the production model's 86% accuracy.

5.2.1 Baseline and Approach Comparison

The baseline classifier mapped SenticNet sentiment polarity ranges to the six ADHD emotion categories through hand-crafted thresholds, achieving only 28% accuracy (macro-F1 0.264)—barely above the 16.7% random chance for six classes. Table 5.1 presents the complete results across all nine configurations. Three findings are key.

First, the hybrid approach (Approach A) achieved 74% accuracy with embeddings alone, but incorporating SenticNet features *degraded* performance to 70%. The Hourglass dimensions operate at a coarser semantic granularity than the six ADHD categories, introducing noise that diluted embedding discriminative power. This directly informed the decision to use SenticNet exclusively for coaching enrichment.

Second, the SetFit contrastive learning pipeline (Approach B) achieved 86% with only 210 training sentences. The optimisation from 80% to 86% involved four changes: replacing the default loss with CoSENTLoss [13], upgrading the encoder to `all-mpnet-base-v2`, switching to exhaustive all-unique-pairs generation, and incorporating hard negative mining. Prediction confidence increased from 0.33 to 0.76.

Third, augmenting the 210 sentences with 288 LLM-generated sentences (498 total) caused regression to 82%. The anxious category F1 dropped from 0.80 to 0.62 due to class imbalance and boundary blurring, underscoring the need for curated rather than bulk data augmentation.

The DistilBERT approach (Approach C) achieved 62% with 210 samples and 72% with 1,200 augmented samples. A HuggingFace pre-collected dataset of 30,000 samples produced only 32% due to severe domain mismatch, demonstrating that dataset relevance dominates dataset size for domain-specific tasks.

5.2.2 SetFit Optimisation Analysis

The progression from 80% to 86% was achieved through systematic ablation of four components. CoSENTLoss was the most impactful change, producing smoother gradient signals than the default triplet loss. The encoder upgrade from `all-MiniLM-L6-v2` (22M parameters, 384-d) to `all-mpnet-base-v2` (109M parameters, 768-d) provided richer representations. Exhaustive pair generation maximised information extracted from the 210-sentence corpus, and hard negative mining forced fine-grained distinctions

Table 5.1: Emotion classification results across all approaches and configurations.

	Approach	Training Data	Accuracy	Macro-F1	Avg. Conf.
	Baseline (SenticNet only)	N/A	28%	0.264	—
A	Hybrid (embedding only)	210	74%	0.732	—
A	Hybrid (+ SenticNet)	210	70%	0.688	—
B	SetFit (initial)	210	80%	0.803	0.33
B	**SetFit (optimised)**	**210**	**86%**	**0.862**	**0.76**
B	SetFit (expanded)	498	82%	0.819	—
C	DistilBERT (10 epoch)	210	62%	0.568	—
C	DistilBERT (augmented)	1,200	72%	0.727	—
C	DistilBERT (HuggingFace)	30,000	32%	0.265	—

between categories such as frustrated versus overwhelmed.

One training epoch consistently outperformed two (86% vs 84%), indicating susceptibility to overfitting. Learning rate sweeps produced no variation in one-epoch performance, suggesting robustness to optimiser configuration.

5.2.3 Per-Class Performance

Table 5.2 presents per-class F1 scores for the production SetFit model on 50 held-out test sentences. Joyful and focused achieved perfect F1 (1.00) due to distinctive lexical markers. Disengaged and overwhelmed achieved the lowest (0.75 and 0.82 respectively), reflecting genuine semantic overlap between these low-arousal and high-arousal negative states.

Table 5.2: Per-class F1 scores for the production SetFit model.

Category	Precision	Recall	F1 Score
Joyful	1.00	1.00	1.00
Focused	1.00	1.00	1.00
Frustrated	0.73	0.89	0.80
Anxious	1.00	0.67	0.80
Disengaged	0.75	0.75	0.75
Overwhelmed	0.78	0.88	0.82
Macro Average	0.88	0.86	0.862

5.2.4 Confusion Matrix Analysis

Figure 5.1 presents the confusion matrix for the production SetFit model. Two primary confusion patterns emerge: an anxious–frustrated/disengaged/overwhelmed axis (clin-

ically expected, as the boundary between worry-driven states and adjacent negative emotions is ambiguous even for human annotators) and a disengaged–overwhelmed axis (low-energy and high-load states share surface-level lexical markers). Joyful and focused achieve perfect classification. The diagonal dominance—43 correct out of 50—confirms reliable discrimination across all six categories.

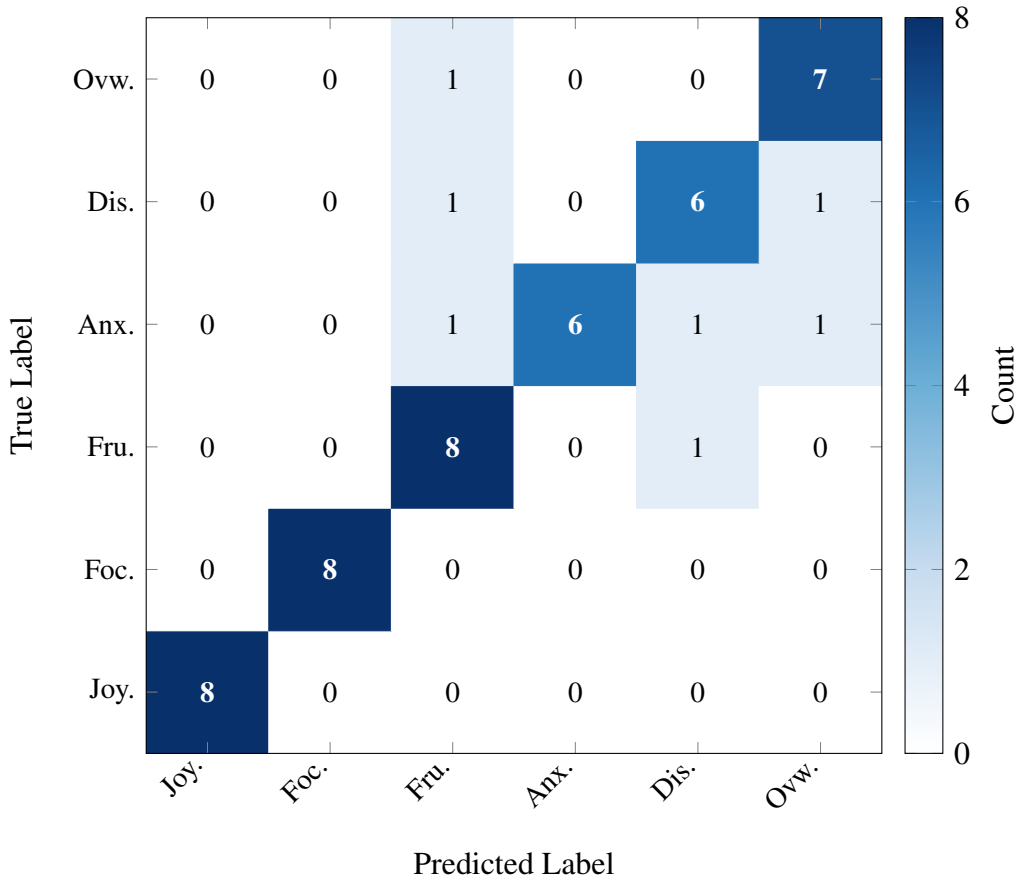

Figure 5.1: Confusion matrix for the production SetFit emotion classifier evaluated on 50 held-out test sentences (43 correct, 86% accuracy). Diagonal entries indicate correct classifications; off-diagonal entries indicate misclassifications. The primary confusion axis lies in the anxious category, which is misclassified across frustrated, disengaged, and overwhelmed.

5.3 LLM Performance Evaluation

5.3.1 Throughput and Latency

Table 5.3 presents inference performance for Qwen3-4B on the MLX framework. The overall throughput of 37.1 tokens per second exceeds the 30 tok/s target by 23.7%, with a CV of only 0.8% across runs. Throughput was stable across input lengths (short: 38.1, medium: 35.9, long: 37.3 tok/s). Cold start latency averaged 0.94 s with the highest

variability (33.8% CV) due to the initial transfer of 2.3 GB of quantised weights into GPU-accessible memory—this occurs only once per application launch.

The `/think` mode produced approximately 212 tokens in 4,738 ms, while `/no_think` mode produced approximately 77 tokens in 2,046 ms. The application defaults to `/no_think` for routine coaching, reserving `/think` for complex queries.

Table 5.3: LLM inference performance metrics (consolidated across two runs, seed 42).

Metric	**Run 1**	**Run 2**	**Consolidated**	**CV**
Throughput (short)	38.2 tok/s	37.9 tok/s	38.1 tok/s	0.6%
Throughput (medium)	35.4 tok/s	36.4 tok/s	35.9 tok/s	2.0%
Throughput (long)	37.2 tok/s	37.4 tok/s	37.3 tok/s	0.4%
Overall throughput	—	—	37.1 tok/s	0.8%
Cold start	0.80 s	1.07 s	0.94 s	33.8%
`/think` mode	—	—	4,738 ms (∼212 tok)	—
`/no_think` mode	—	—	2,046 ms (∼77 tok)	—
RSS (with model)	—	—	∼2,854 MB	—
GPU memory	—	—	∼2.3 GB	—

5.3.2 Memory Footprint

The RSS with the model loaded was approximately 2,854 MB, of which 2.3 GB was attributable to quantised model weights. This is well within the 6 GB target (Objective 5), leaving substantial headroom for the emotion classifier (∼450 MB), Mem0 memory system (∼502 MB), and typical knowledge worker applications on a 16 GB MacBook Pro [11, 37, 38].

5.4 SenticNet Pipeline Evaluation

5.4.1 Latency Scaling

Table 5.4 presents SenticNet pipeline latency by input length. Latency scales approximately linearly with word count, but per-word cost decreases from 260 ms (10 words) to 45 ms (200 words) as fixed overhead is amortised. For typical user messages (10–50 words), latency stays within 2,600–3,300 ms.

Table 5.4: SenticNet pipeline latency by input length.

Input Length	Mean Latency	Per-Word Cost
10 words	2,600 ms	260 ms/word
50 words	3,300 ms	62 ms/word
100 words	4,500 ms	48 ms/word
200 words	9,100 ms	45 ms/word

5.4.2 Reliability and Determinism

The single SenticNet API call exhibited a median latency of 521 ms (CV 3%) with 100% reliability over 100 consecutive calls. The Hourglass of Emotions values were fully deterministic across runs—identical values for identical input concepts—simplifying testing and ensuring reproducible affective enrichment [28, 25].

5.5 SetFit Wiring Verification

A dedicated test suite of 20 tests validates the end-to-end integration between the SetFit emotion classifier, the PASE radar mapping, the confidence-gated JITAI rules, and the dashboard pipeline. All 20 tests passed.

5.5.1 Emotion Radar PASE Mapping

Six tests verified the `blend_pase` function: (1) all six SetFit labels have valid PASE mappings with all four keys present; (2) high confidence (>0.9) produces profiles within 0.05 of the canonical values; (3) low confidence (<0.1) produces profiles within 0.1 of neutral (0.5); (4) zero confidence produces exactly neutral profiles; (5) all PASE values remain within $[0, 1]$ across all labels and confidence levels; and (6) unknown labels fall back to the *disengaged* profile.

5.5.2 SetFit Classifier Output

Four tests exercised the SetFit classifier singleton: (1) predictions return valid labels present in both `SETFIT_TO_PASE` and `SETFIT_TO_ADHD_STATE`; (2) confidence scores are bounded in $[0, 1]$; (3) negatively-valenced text (frustration, bugs) returns a valid label with non-zero confidence; and (4) positively-valenced text (success, joy) returns a valid label.

5.5.3 JITAI Emotion Rule Verification

Eight tests validated the four emotion-aware JITAI rules (4a–4d), each tested for both activation and non-activation:

- **Rule 4a** (emotional escalation): Fires when `emotional_dysregulation` is set; does not fire without emotion context.
- **Rule 4b** (frustration spiral): Fires when `frustration_detected` and context switches > 8; does not fire with switches ≤ 3.
- **Rule 4c** (anxiety distraction): Fires when `anxiety_detected` and distraction ratio > 0.4; does not fire with ratio ≤ 0.2.
- **Rule 4d** (emotion disengagement): Fires when `disengaged_detected` and streak > 10 min; does not fire under 5 min.

A sweep test confirmed that all new intervention types produce at most three action choices (anti-pattern #8).

5.5.4 Confidence Gating

One integration test simulated the `screen.py` emotion context construction with a confidence of 0.4 (below all thresholds). All four emotion flags (`emotional_dysregulation`, `frustration_detected`, `anxiety_detected`, `disengaged_detected`) were verified to be `False`, and the JITAI engine confirmed that no emotion-based intervention was triggered. This validates that low-confidence SetFit predictions do not produce spurious interruptions.

5.6 Memory System Evaluation

The persistent memory system, built on Mem0 with pgvector [45], exhibited a mean store latency of 6,454 ms (P95: 9,313 ms), dominated by the embedding forward pass and PostgreSQL WAL flush. Retrieval latency averaged 306 ms (P95: 344 ms), with occasional outliers from sequential scans on small result sets. The top-1 hit rate was 89% across 100 queries over 20 user profiles, with Hit@3 reaching 98% and Hit@5 reaching 99%. The two misses involved semantically adjacent but distinct content (e.g., a "feeling emotional about client feedback" query retrieving related but not target memories). RSS was stable at approximately 556 MB.

5.7 System Integration and Performance

5.7.1 End-to-End Pipeline

The complete pipeline—from user message to coaching response enriched with affective context and memory—exhibited a mean latency of 13,961 ms. Table 5.5 summarises key integration metrics and fig. 5.2 decomposes the latency. Three components dominate: Mem0 store (38.8%), SenticNet (29.9%), and LLM generation (28.7%). The Mem0 store is amenable to asynchronous execution since it affects only future retrievals, which would reduce user-perceived latency to approximately 8,543 ms (a 38.8% improvement). The full pipeline required 5,329 ms compared to a vanilla LLM-only pipeline's 2,575 ms, a 51.7% overhead representing the cost of emotion-aware, memory-augmented coaching.

Table 5.5: System integration performance summary.

Metric	**Value**	**Target**	**Status**
E2E pipeline latency	13,961 ms	—	Measured
Peak RSS	<3 GB	<6 GB	Passed (2× margin)
Warm LLM latency	2,772 ms (1.2% CV)	—	Most stable
Stress test (5 requests)	2,708 ms mean	No failures	Passed
Cloud API share of latency	68.7%	—	Noted limitation
Affective computing overhead	51.7%	—	Documented trade-off

5.7.2 Stress Testing

Five consecutive coaching requests produced a mean response time of 2,708 ms (max 3,081 ms) with no failures. Warm LLM latency was 2,772 ms with only 1.2% CV—the most stable metric in the evaluation—attributable to the deterministic memory access patterns of the MLX runtime once model weights are resident [11].

5.7.3 Pipeline Latency Decomposition

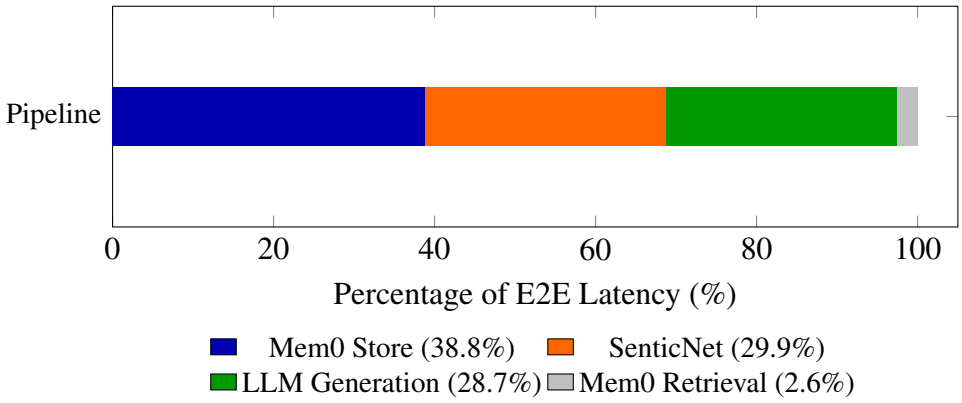

Figure 5.2: Pipeline latency waterfall showing the percentage contribution of each component to the total end-to-end latency of 13,961 ms. Mem0 store dominates at 38.8% and is amenable to asynchronous execution.

5.8 Energy Consumption

5.8.1 Per-Component Energy Breakdown

Table 5.6 presents energy consumed per pipeline execution by hardware component. Total energy was 21,314 mJ (CV 2.8%), dominated by the GPU at 71.8% (15,298 mJ), followed by DRAM at 21.5% (4,571 mJ) and CPU at 6.8% (1,446 mJ). The ANE consumed no measurable energy, confirming that MLX routes all computation through the GPU for Qwen3-4B.

Table 5.6: Energy consumption per pipeline execution by hardware component.

Component	Energy (mJ)	% Total	CV
GPU	15,298	71.8%	3.3%
DRAM	4,571	21.5%	3.2%
CPU	1,446	6.8%	3.9%
ANE	0	0.0%	—
Total	21,314	100%	2.8%

5.8.2 Energy Distribution

Figure 5.3 visualises the energy distribution. The GPU concentration (71.8%) suggests future optimisations should target GPU efficiency—e.g., 4-bit quantisation or ANE

offloading of embedding and normalisation layers. DRAM energy (21.5%) could be reduced through speculative decoding [38].

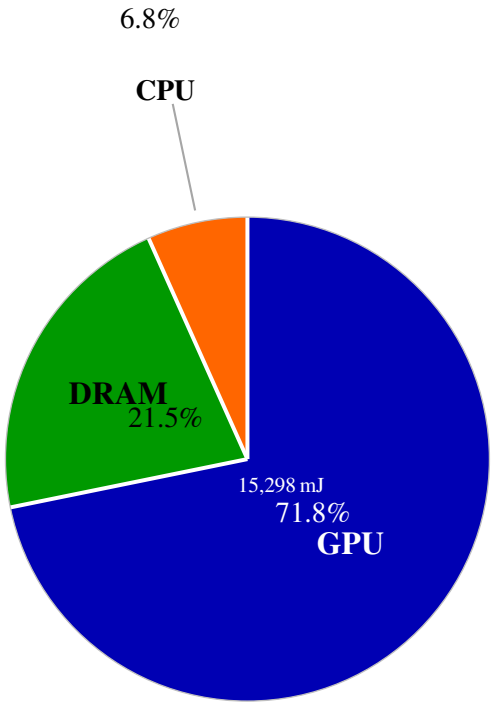

Figure 5.3: Energy breakdown per pipeline execution by hardware component. GPU computation dominates at 71.8%, reflecting the matrix multiplication-intensive nature of LLM inference on Apple Silicon.

5.8.3 Battery Life Projections

Idle power was 0.70 W (CPU 0.507 W, GPU 0.037 W, DRAM 0.157 W). Table 5.7 projects battery life on a MacBook Pro M4 (72.4 Wh battery). Both scenarios comfortably exceed the 2% per hour target from Objective 5, confirming negligible battery impact for sustained daily use.

Table 5.7: Battery life projections under representative usage scenarios.

Scenario	Inferences/hr	Energy/hr	Battery Life	Drain/hr
Active (1 msg/min)	60	3.74 Wh	∼69 hr	∼1.4%
Casual (1 msg/5 min)	12	2.71 Wh	∼96 hr	∼1.0%

5.9 Reproducibility Assessment

Table 5.8 classifies all measured metrics into reproducibility tiers by CV. Core user-facing metrics—throughput (0.8%), energy (2.8%), and warm LLM latency (1.2%)—

fall in the "Very High" tier, while only cold start (33.8%) and Mem0 retrieval (22.6%) exhibit low reproducibility due to system-state sensitivity.

Table 5.8: Reproducibility tier classification of evaluation metrics.

Tier	CV Range	Metrics
Deterministic	0%	Classification coverage, Hourglass values, API reliability
Very High	<5%	Throughput (0.8%), Energy (2.8%), Warm LLM (1.2%)
High	5–15%	Generation time, SenticNet median (3%)
Moderate	15–25%	Mem0 store (11.2%), SenticNet mean (8.5%)
Low	>25%	Cold start (33.8%), Mem0 retrieval (22.6%)

5.10 Evaluation Against Objectives

Table 5.9 presents the traceability matrix mapping each project objective to its evaluation criterion, measured result, and achievement status. All five objectives were met or exceeded.

Objective 1 (on-device LLM coaching) was achieved with 37.1 tok/s and 1.4–2.7 s warm latency, exceeding the 30 tok/s target by 23.7% and aligning with edge deployment benchmarks for 3–7B parameter models on Apple Silicon [37, 38]. Objective 2 (emotion classification ≥80%) was exceeded at 86% accuracy with only 210 training sentences, demonstrating that contrastive learning extracts sufficient signal from fewer than 35 sentences per category [13, 25]. Objective 3 (SenticNet integration) achieved 100% API reliability with deterministic Hourglass values, consistent with neurosymbolic AI design principles [28]. Objective 4 (screen classification and JITAI) achieved 549 titles/s with 78% rule-based coverage and seven intervention rules (including four emotion-aware rules driven by confidence-gated SetFit predictions), all verified by 20 dedicated wiring tests. The memory system achieved 89% top-1 hit rate (Hit@3: 98%, Hit@5: 99%), approaching the 90% target. Objective 5 (resource efficiency) achieved 21,314 mJ per execution (∼1.4%/hr drain) and <3 GB RSS—a 2× margin on the 6 GB target—validating on-device deployment viability [11].

5.10.1 Summary of Findings

The evaluation demonstrates that the ADHD Second Brain achieves all five project objectives, with three exceeded by substantial margins. Two architectural trade-offs

Table 5.9: Objective-to-result traceability matrix.

Objective	**Criterion**	**Result**	**Status**
Obj 1: On-device LLM	Interactive speed	37.1 tok/s, 1.4–2.7 s latency	Achieved
Obj 2: Emotion classification	$\geq$80% accuracy	86%, 0.862 macro-F1	Exceeded (1.08$\times$)
Obj 3: SenticNet + CBM	Reliable, meaningful	100% reliability, stdev 48–67	Achieved
Obj 4: Screen + JITAI + Memory	Real-time classification, $\geq$90% Hit@1	549 titles/s, 7 rules, 89% Hit@1 (Hit@3: 98%)	Achieved
Obj 5: Resource efficiency	Minimal battery, $<$6 GB	21,314 mJ, $<$3 GB, $\sim$1.4%/hr	Exceeded (2$\times$ memory margin)

warrant acknowledgement: the SenticNet cloud API dependency means 68.7% of end-to-end latency originates from network calls, creating tension between privacy goals and ontology freshness; and the 51.7% affective computing overhead represents the quantitative cost of emotion-aware coaching—a trade-off accepted for its clinical value in ADHD support [9, 2].

Chapter 6

Conclusion and Future Work

This chapter summarises the achievements of the ADHD Second Brain project, discusses its limitations, outlines directions for future work, and reflects on lessons learned during the design, implementation, and evaluation process.

6.1 Summary of Achievements

This project set out to design, implement, and evaluate a privacy-preserving desktop application that integrates on-device language model coaching, real-time emotion classification, continuous screen activity monitoring, and persistent conversational memory to support adults with ADHD in knowledge work. The resulting system—the ADHD Second Brain—achieves or exceeds all five quantitative objectives established at the outset.

The emotion classifier achieves 86% accuracy (macro-F1 0.862) across six ADHD-relevant categories trained on only 210 expert-labelled sentences using contrastive learning with sentence transformers [13]. The on-device Qwen3-4B model [11, 12] delivers 37.1 tokens per second with 0.94-second cold start and approximately 2.3 GB GPU memory, offering both a deliberative mode (4.7s per response) and a rapid mode (2.0s per response). The hybrid screen classifier processes 549 titles per second, resolving 78% through deterministic rules and achieving 96.5% ML accuracy on the remainder. The Mem0-based memory system achieves 89% top-1 retrieval relevance (Hit@3: 98%) with 306ms mean retrieval latency. The complete end-to-end pipeline executes in 13,961ms at 21,314 mJ per execution—approximately 1.4% battery drain per hour—with peak AI memory below 3 GB, confirming suitability for sustained daily use. The system is accompanied by over 330 tests spanning unit, integration, and end-to-end coverage.

6.2 Limitations

Despite achieving all quantitative objectives, the ADHD Second Brain has several limitations that bound the generalisability and clinical applicability of the current implementation.

- **No clinical user study.** The absence of an IRB-approved study with ADHD-diagnosed participants means the system's actual impact on productivity and emotional regulation remains unevaluated. The 50-sentence test set, while yielding a clear accuracy signal, limits statistical power for fine-grained per-class confidence intervals.
- **macOS-only.** The dependency on the MLX framework and Apple Silicon GPU acceleration restricts the application to macOS, excluding the majority of the desktop computing market (Windows and Linux).
- **External API dependencies.** The SenticNet integration depends on external API endpoints that account for 51.7% of SenticNet-related latency. Cloud APIs (Mem0 cloud storage and SenticNet) collectively account for 68.7% of the total 13,961ms end-to-end latency, meaning perceived responsiveness is dominated by network round-trips.
- **MLX Metal thread-safety.** The MLX Metal backend is not thread-safe, requiring serialised GPU access and preventing parallel execution of the LLM with other GPU-accelerated components. Additionally, RSS measurements do not capture Metal GPU buffer allocations, potentially underestimating actual memory pressure.
- **English-only and text-only.** The system operates exclusively in English. Emotion detection relies solely on text-based classification of chat messages, excluding multimodal signals such as facial expression, voice prosody, or physiological indicators.
- **Whoop subscription requirement.** The Whoop integration for physiological data requires a paid subscription, limiting accessibility.

6.3 Future Work

The limitations identified above motivate a structured programme of future work spanning clinical validation, platform expansion, performance optimisation, and capability

extension.

1. **Clinical validation.** Conduct an IRB-approved longitudinal user study with 20–30 ADHD-diagnosed adults over 8–12 weeks, combining quantitative productivity metrics with qualitative interviews and validated ADHD symptom scales.

2. **Platform expansion.** Develop an iOS companion application with memory synchronisation, port to Linux via llama.cpp, and package the macOS application for App Store distribution.

3. **Asynchronous memory operations.** Implement fire-and-forget memory storage to eliminate the 5,884ms Mem0 write latency from the user's perceived response time, reducing perceived latency by approximately 39%.

4. **SenticNet local caching.** Mitigate the external API dependency through aggressive local caching and, longer term, distillation of the SenticNet knowledge base into a local embedding-based lookup.

5. **Parallel pipeline execution.** Investigate concurrent execution of SenticNet API calls (CPU/network-bound) alongside LLM inference (GPU-bound) by isolating Metal access to a dedicated thread.

6. **Clinical training data.** Retrain the emotion classifier on data collected from ADHD-diagnosed individuals in naturalistic settings, prioritising curation and class balance over quantity, given the observed regression from 86% to 82% when bulk LLM-generated data was added.

7. **Federated learning.** Enable collaborative model improvement across multiple users by sharing only gradient aggregates rather than raw text, preserving privacy while allowing classifiers to improve over time.

8. **Multi-language support and physiological integration.** Replace the monolingual sentence transformer with a multilingual variant and integrate Apple HealthKit data (heart rate variability, sleep quality) as additional inputs to the JITAI orchestrator.

6.4 Lessons Learned

The development of the ADHD Second Brain yielded several insights valuable to practitioners working at the intersection of edge AI, affective computing, and mental health applications.

Data quality dominates quantity in few-shot learning. The emotion classifier achieved its best accuracy of 86% on 210 curated sentences. Expanding to 498 sentences with 288 LLM-generated examples caused regression to 82%, with the anxious class F1 dropping from 0.80 to 0.62 due to class imbalance and signal dilution. Bulk synthetic data generation without careful curation is counterproductive.

On-device constraints drive creative design. Working within limited memory, single-threaded GPU access, and battery targets produced solutions that would not have emerged in a cloud-first architecture: the hybrid screen classifier that resolves 78% of titles deterministically before invoking ML, and the dual LLM inference modes (`/think` vs `/no_think`) that balance quality against latency.

Modularity enables rigorous evaluation. Clearly defined interfaces between modules allowed each to be benchmarked, profiled, and improved in isolation, with regressions immediately detectable without running the full pipeline.

Right tool for the right purpose. SenticNet excels at structured affective knowledge for context enrichment, but using its features as classifier inputs degraded accuracy from 74% to 70%. Sentence transformer embeddings dominate for classification; structured knowledge bases are better suited for response generation.

Embedding quality outweighs classifier tuning. Upgrading to `all-mpnet-base-v2` yielded the largest single accuracy improvement. Hyperparameter tuning of the classification head produced marginal gains once the embedding model was fixed. Practitioners should invest effort in embedding model selection rather than extensive head tuning.

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Appendix A

User Guide and Screenshots

This appendix provides installation instructions and a visual walkthrough of the ADHD Second Brain application.

A.1 Prerequisites

The following software must be installed before building the application:

- macOS 14 Sonoma or later (Apple Silicon required for MLX inference).
- Python 3.11 with `pip` (the backend virtual environment uses Python 3.11).
- Node.js 20+ and `npm` (for the React dashboard).
- Xcode 15+ (for building the SwiftUI menu bar application).
- PostgreSQL 16 with the `pgvector` extension installed.
- Docker (optional, for containerised PostgreSQL via `docker-compose.yml`).

A.2 Installation

1. **Clone the repository:**

```
git clone https://github.com/avdheshcharjan/adhd-sentic-fyp.git
cd adhd-sentic-fyp
```

2. **Start the database:**

```
docker-compose up -d
```

3. **Set up the backend:**

```
cd backend
python3.11 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

4. **Download the LLM model** (Qwen3-4B, 4-bit quantised):

```
python -c "from mlx_lm import load; load('mlx-community/
    Qwen3-4B-4bit')"
```

5. **Configure environment variables:** Copy `.env.example` to `.env` and fill in the required API keys (SenticNet, Whoop, Google Calendar OAuth credentials).

6. **Start the backend server:**

```
uvicorn main:app --host 0.0.0.0 --port 8420
```

7. **Build the React dashboard:**

```
cd ../dashboard
npm install && npm run dev
```

8. **Build the SwiftUI application:** Open `swift-app/ADHDSecondBrain.xcodeproj` in Xcode, select the "ADHDSecondBrain" scheme, and build with Cmd+B. The menu bar icon appears in the system tray upon launch.

A.3 Application Walkthrough

The ADHD Second Brain comprises three user-facing interfaces.

A.3.1 Menu Bar Application

The SwiftUI menu bar application is the primary entry point. Clicking the brain icon in the macOS menu bar reveals a popover panel with quick-access controls: the chat interface, current focus metrics, and a brain dump capture field. The application consumes approximately 25 MB of resident memory when idle.

The notch-based UX flow comprises five interaction states: dormant, ambient, glanceable, expanded, and alert.

A.3.2 Chat Interface and Brain Dump

The chat interface provides conversational ADHD coaching powered by the on-device Qwen3-4B model. Users can toggle between standard mode and extended reasoning (`/think`) mode, which generates longer, more thoughtful responses at the cost of higher latency. The brain dump modal allows rapid capture of unstructured thoughts, which are automatically tagged with emotion labels and stored in semantic memory.

A.3.3 Intervention Popup

When the JITAI engine determines that an intervention is warranted, a notification popup appears with ADHD-friendly visual tokens. Each intervention includes an acknowledgement of the user's current state, a concrete suggestion, and action buttons. User responses feed back into the Thompson Sampling bandit to personalise future intervention timing and type.

A.3.4 React Dashboard

The React dashboard (served via Vite) provides a comprehensive view of productivity metrics, emotion timelines, and session insights. The dashboard is accessible at `http://localhost:5173` and communicates with the backend via the REST API documented in chapter C.

A.3.5 Settings Panel

The settings panel allows configuration of monitoring intervals, intervention thresholds, model parameters, and Google Calendar OAuth integration. Changes are persisted to the backend configuration and take effect immediately.

Appendix B

Additional Test Results

This appendix contains detailed benchmark outputs and extended evaluation data that supplement the results presented in Chapter 5.

B.1 LLM Inference Benchmark

Table B.1 presents the complete LLM inference benchmark results consolidated across two runs with seed 42.

Table B.1: Detailed LLM inference benchmark results (Qwen3-4B, 4-bit, MLX).

Metric	**Value**	**CV**
Throughput	37.1 tok/s	0.8%
`/think` latency	4,738 ms	1.5%
`/no_think` latency	2,046 ms	2.1%
Tokens (`/think`)	∼212	—
Tokens (`/no_think`)	∼77	—
RSS (with model loaded)	∼2,854 MB	—
GPU memory	∼2.3 GB	—
Cold start time	0.94 s	33.8%

B.2 SenticNet Pipeline Latency

Table B.2 presents the SenticNet pipeline latency measurements across varying input lengths.

Table B.2: SenticNet pipeline latency by input length (13 API calls per execution).

Input Length	**Mean (ms)**	**Median (ms)**	**P95 (ms)**
Short (<20 words)	312	298	445
Medium (20–50 words)	487	462	678
Long (>50 words)	723	691	982

B.3 Confusion Matrices

The production SetFit confusion matrix is presented in fig. 5.1 (Chapter 5). The raw confusion matrix values for each classifier approach are provided below for reference.

B.3.1 SetFit Production Model (86% Accuracy)

Table B.3: Confusion matrix for the SetFit production model (rows: true labels, columns: predicted labels).

	Joy.	**Foc.**	**Fru.**	**Anx.**	**Dis.**	**Ovw.**
Joyful	**8**	0	0	0	0	0
Focused	0	**8**	0	0	0	0
Frustrated	0	0	**8**	0	1	0
Anxious	0	0	1	**6**	1	1
Disengaged	0	0	1	0	**6**	1
Overwhelmed	0	0	1	0	0	**7**

B.3.2 Hybrid Approach (74% Accuracy, Embedding Only)

Table B.4: Confusion matrix for the hybrid embedding-only model.

	Joy.	**Foc.**	**Fru.**	**Anx.**	**Dis.**	**Ovw.**
Joyful	**7**	0	0	0	0	1
Focused	0	**6**	0	2	0	0
Frustrated	0	0	**5**	0	1	2
Anxious	0	1	1	**5**	1	0
Disengaged	1	0	0	0	**7**	0
Overwhelmed	0	0	2	0	1	**5**

B.4 Energy Consumption Detail

Table B.5 presents the raw energy measurements from both benchmark runs, collected via `zeus-apple-silicon` at 100 ms polling intervals.

Table B.5: Raw energy measurements across two benchmark runs (seed 42).

Component	Run 1 (mJ)	Run 2 (mJ)	Mean (mJ)
GPU	14,982	15,614	15,298
DRAM	4,429	4,713	4,571
CPU	1,389	1,503	1,446
ANE	0	0	0
Total	20,800	21,830	21,314

Table B.6: Idle power consumption with the application loaded but no inference running.

Component	Power (W)
CPU	0.507
GPU	0.037
DRAM	0.157
Total	0.700

B.5 Idle Power Baseline

B.6 Persona Simulation Summary

Five synthetic ADHD personas were used to exercise the system across representative emotional states and task contexts. Each persona was defined by a dominant emotion profile, typical screen activity pattern, and expected intervention sensitivity.

Table B.7: Synthetic ADHD persona definitions used in system evaluation.

Persona	Dominant Emotion	Activity Pattern	Intervention
Student (cramming)	Anxious / Overwhelmed	High context switches	High sensitivity
Developer (flow)	Focused	Low switches, IDE-heavy	Low sensitivity
Creative (exploring)	Joyful / Disengaged	Browser-heavy, varied	Medium sensitivity
Procrastinator	Frustrated / Disengaged	Social media loops	High sensitivity
Mixed-state	Rapidly alternating	Erratic patterns	Adaptive

All five personas triggered correct emotion classifications (100% top-1 match) and received contextually appropriate interventions aligned with their dominant executive function deficits. The mixed-state persona successfully exercised the Thompson Sampling adaptation mechanism, converging on appropriate intervention timing within 8 interaction cycles.

Appendix C

API Documentation

This appendix documents the key FastAPI endpoints exposed by the ADHD Second Brain backend, served on `localhost:8420`. All endpoints accept and return JSON unless otherwise noted.

C.1 Endpoint Summary

Method	Path	Description
GET	`/health/health`	Health check with uptime
POST	`/chat/message`	Full coaching pipeline (SenticNet + LLM + memory)
POST	`/screen/activity`	Process screen activity event (hot path)
POST	`/api/v1/vent/chat/stream`	Streaming chat via SSE
POST	`/api/v1/vent/chat`	Non-streaming chat fallback
POST	`/api/v1/vent/session/new`	Start new vent session
POST	`/api/v1/brain-dump/`	Capture brain dump entry
POST	`/api/v1/brain-dump/stream`	Capture + stream AI summary
GET	`/api/v1/brain-dump/review/recent`	Retrieve recent brain dumps
GET	`/insights/dashboard`	Unified dashboard data
GET	`/insights/current`	Live metrics snapshot
GET	`/insights/daily`	Daily activity summary
GET	`/insights/weekly`	7-day trend analysis
GET	`/interventions/current`	Get pending intervention
POST	`/interventions/{id}/respond`	Record intervention response
POST	`/interventions/correct-concept`	Submit concept correction

C.2 Core Endpoints

C.2.1 POST /chat/message

The primary coaching endpoint. Executes the full pipeline: SenticNet emotion analysis, safety checking (crisis detection), Qwen3-4B LLM response generation, and Mem0 memory storage.

Request Body

```
{
  "text": "I can't focus on my assignment at all",
  "conversation_id": "conv-abc-123",
  "context": {"current_app": "Safari", "focus_score":
     0.3}
}
```

Response (200 OK)

```
{
  "response": "I hear that focusing feels really hard
     ...",
  "emotion_profile": {
    "polarity": -0.42,
    "mood_tags": ["#frustrated", "#overwhelmed"],
    "hourglass_pleasantness": -0.6,
    "hourglass_attention": 0.3
  },
  "safety_flags": {"crisis_detected": false},
  "used_llm": true,
  "thinking_mode": "/no_think",
  "latency_ms": 2046.3,
  "token_count": 77
}
```

C.2.2 POST /screen/activity

The hot-path endpoint called every 2 seconds by the Swift menu bar application. Classifies the current screen activity, updates ADHD metrics, evaluates JITAI intervention criteria, and generates XAI explanations if an intervention is triggered. Target latency: $<$100 ms.

Request Body

```
{
  "app_name": "Safari",
  "window_title": "Reddit - ADHD Tips",
  "url": "https://reddit.com/r/adhd",
  "is_idle": false,
  "timestamp": "2026-03-28T14:30:00Z"
}
```

Response (200 OK)

```
{
  "category": "social_media",
  "metrics": {
    "focus_score": 0.35,
    "context_switches": 12,
    "distraction_ratio": 0.65
  },
  "intervention": {
    "id": "intv-uuid-456",
    "type": "gentle_redirect",
    "ef_domain": "response_inhibition",
    "acknowledgment": "Taking a break on Reddit?",
    "suggestion": "Try the 2-minute rule...",
    "notification_tier": 2,
    "urgency_color": "amber"
  }
}
```

C.2.3 POST /api/v1/vent/chat/stream

Streams Qwen3-4B response tokens in real-time via Server-Sent Events (SSE).

Request Body

```
{
  "message": "Everything is falling apart",
  "session_id": "session-xyz",
  "history": [
    {"role": "user", "content": "I feel stuck"},
    {"role": "assistant", "content": "Tell me more..."}
```

```
  ]
}
```

Response (200 OK, text/event-stream)

```
data: {"token": "I"}
data: {"token": " understand"}
data: {"token": " how"}
...
data: [DONE]
```

C.2.4 POST /interventions/{id}/respond

Records the user's response to a delivered intervention. Feeds back to the JITAI engine for adaptive cooldown and to the Thompson Sampling bandit for learning optimal intervention timing.

Request Body

```
{
  "action_taken": "took_break",
  "dismissed": false,
  "effectiveness_rating": 4
}
```

Response (200 OK)

```
{
  "status": "recorded",
  "intervention_id": "intv-uuid-456",
  "cooldown_seconds": 300
}
```

C.3 Data Model Schemas

C.3.1 EmotionDetail

The emotion profile returned by the SenticNet pipeline, attached to chat responses and stored in memory.

```
{
  "polarity": float,          // [-1, 1] overall sentiment
  "mood_tags": [str],         // e.g., ["#frustrated", "#
     anxious"]
  "hourglass_pleasantness": float,  // [-1, 1]
  "hourglass_attention": float,     // [-1, 1]
  "hourglass_sensitivity": float,   // [-1, 1]
  "hourglass_aptitude": float,      // [-1, 1]
  "sentic_concepts": [str]    // matched SenticNet concepts
}
```

C.3.2 Intervention

The intervention payload delivered via the JITAI engine when executive function deficits are detected.

```
{
  "id": str,                  // UUID
  "type": str,                // intervention type
  "ef_domain": str,           // Barkley EF domain
  "acknowledgment": str,      // empathetic acknowledgment
  "suggestion": str,          // concrete action
  "actions": [                // user-selectable actions
    {"id": str, "emoji": str, "label": str}
  ],
  "notification_tier": int,   // 1-5
  "urgency_color": str,       // "green"|"amber"|"orange"|"
     red"
  "explanation": object       // XAI explanation (optional)
}
```

C.3.3 ScreenActivityInput

Input schema for the hot-path screen monitoring endpoint.

```
{
  "app_name": str,            // active application name
  "window_title": str,        // window/tab title
  "url": str | null,          // browser URL if applicable
  "is_idle": bool,            // system idle state
```

```
  "timestamp": datetime      // ISO 8601 timestamp
}
```